

Sharp optimality of sparse group LASSO and SLOPE under weaker assumptions

Zhifan Li¹ , Yanhang Zhang²  and Jianxin Yin³ 

¹*School of Statistics and Mathematics, Zhongnan University of Economics and Law*, e-mail: demarcuslee@zuel.edu.cn

²*Yau Mathematical Sciences Center, Tsinghua University*, e-mail: zhangyh98@tsinghua.edu.cn

³*Center for Applied Statistics and School of Statistics, Renmin University of China*, e-mail: jyin@ruc.edu.cn

Abstract: This paper introduces a rigorous approach to establish the sharp minimax optimalities of both sparse group LASSO and SLOPE, where parameters exhibit both element-wise and group-wise sparsity. We derive an original upper bound for Gaussian complexity based on a weighted $\ell_1 + \ell_{1,2}$ norm. Based on this result, we establish the sharp optimal results for both sparse group LASSO and SLOPE under restricted eigenvalue (RE) type conditions. To the best of our knowledge, the assumptions employed in this paper are the weakest among existing analyses. We find the astonishing result that the intersection of sparse restricted eigenvalue cones for element-wise sparsity and for group-wise sparsity corresponds to the solution of weighted $\ell_1 + \ell_{1,2}$ penalty. Furthermore, based on the proof techniques, we can provide the optimal sample complexity under the framework of random design, where both weak moment assumption and sub-Gaussian random design are considered. Numerical experiments also confirm that, under the double sparsity setting, our algorithm achieves superior performance compared to methods based on only single type of sparsity assumption. Our analysis technique can also be extended to the theoretical analysis of other general structured sparsity problems.

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1. Introduction

1.1. Background

Over the past 30 years, empirical evidence has shown that real predictors in high-dimensional settings can exhibit intricate structural patterns. For example, covariates often exhibit certain group structures where only a few groups contribute significantly to the response, an assumption known as group sparsity [47, 19, 30, 51]. In particular, groups of variables are either entirely included in the model or excluded altogether, a selection approach commonly referred to as “all-in or all-out”.

Beyond group sparsity, double sparse structure considers sparsity within each group [15, 40, 9]. It is further assumed that within each selected group, only a small number of its constituent variables are relevant to the response. Double sparse structure has a wide range of real-world applications, including genetic data analysis [39, 20], climate data analysis [50], as well as Alzheimer’s disease diagnosis [29, 35].

1.2. Model description

We consider the linear regression problem, which is stated as

$$y = X\beta^* + \xi, \quad (1)$$

where $y \in \mathbb{R}^n$ is the response variable, $X \in \mathbb{R}^{n \times p}$ is the design matrix, $\beta^* \in \mathbb{R}^p$ is the underlying coefficient and $\xi \in \mathbb{R}^n$ is the error term in which each entry is drawn independently from $\mathcal{N}(0, \sigma^2)$.

In our settings, the p variables can be divided into m pre-known non-overlapping groups. In particular, $G_j \subseteq [p]$ represents the index set of the j -th group and $\cup_{j \in [m]} G_j = [p]$. For simplification, we assume that each group has the same number of variables, that is, $|G_j| = d$ for each $j \in [m]$. In cases where group sizes differ, we can pad smaller groups with zero coefficients and white noise designs to match the same largest group size without affecting the theoretical results. We say that β is (s, s_0) -sparse if

$$\|\beta\|_{0,2} \triangleq \sum_{j=1}^m \mathbf{I}(\|\beta_{G_j}\|_2 \neq 0) \leq s \quad \text{and} \quad \|\beta\|_0 \triangleq \sum_{i=1}^p \mathbf{I}(\beta_i \neq 0) \leq ss_0.$$

Here, s and ss_0 are two positive constants that control the group-wise and element-wise sparsity, respectively. Specifically, s directly imposes a sparsity constraint at the group level, and s_0 can be interpreted as the average sparsity per group in the support groups. $\mathcal{DS}(s, s_0)$ denotes the double sparse parameter space:

$$\mathcal{DS}(s, s_0) \triangleq \{\beta \in \mathbb{R}^p : \|\beta\|_{0,2} \leq s, \|\beta\|_0 \leq ss_0\}.$$

Given parameter space $\mathcal{DS}(s, s_0)$, we can establish the minimax framework as follows:

$$\inf_{\hat{\beta}} \sup_{\beta^* \in \mathcal{DS}(s, s_0)} \mathbf{E} \|\hat{\beta} - \beta^*\|_2^2.$$

In the previous work, [9, 27] have derived the minimax lower bound $\frac{\sigma^2}{n}(ss_0 \log(\frac{d}{s_0}) + s \log(\frac{m}{s}))$ under some regular condition. In this paper, we will establish the same lower bound under weaker assumptions. We next introduce the sparse group LASSO and the sparse group SLOPE:

1.2.1. Sparse group LASSO

To solve the regression problem under double sparsity, a widely studied approach is the sparse group LASSO [15, 40, 9]. The minimization problem of sparse group LASSO is

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \left(\|y - X\beta\|_n^2 + \lambda \|\beta\|_1 + \lambda_g \|\beta\|_{1,2} \right), \quad (2)$$

where $\lambda, \lambda_g > 0$ are the tuning parameters. Here $\|\beta\|_1$ and $\|\beta\|_{1,2}$ correspond to the penalties of the ordinary LASSO [42] and group LASSO [47], respectively. The specific forms of these norms are defined as

$$\|\beta\|_1 = \sum_{i=1}^p |\beta_i| \quad \text{and} \quad \|\beta\|_{1,2} = \sum_{j=1}^m \|\beta_{G_j}\|_2.$$

When the estimation error of the estimator $\hat{\beta}$ matches the above minimax lower bound, we state that the estimator achieves the minimax optimality. To achieve the minimax optimality of $\hat{\beta}$, the relationship between λ and λ_g in (2) should satisfy $\lambda_g \asymp \sqrt{s_0}\lambda$ [9]. Hence, the penalty term of sparse group LASSO can be written as

$$\lambda (\|\beta\|_1 + \sqrt{s_0}\|\beta\|_{1,2}).$$

1.2.2. Sparse group SLOPE

Similar to the idea of SLOPE [6], we define the sorted $\ell_1 + \ell_{1,2}$ penalty as follows: the sparse group SLOPE estimator $\hat{\beta}$ is defined as a solution of the minimization problem

$$\hat{\beta} \in \operatorname{argmin}_{\beta \in \mathbb{R}^p} \left(\|y - X\beta\|_n^2 + \|\beta\|_* \right). \quad (3)$$

Specifically,

$$\|\beta\|_* = \sum_{j=1}^m \lambda_j \|\beta_{G(j)}\|_2 + \sum_{i=1}^p \tilde{\lambda}_i |\beta_{(i)}|,$$

where $|\beta_{(i)}|$ denotes i -th largest absolute value of β and $\|\beta_{G(j)}\|_2$ denotes the j -th largest ℓ_2 -norm of $\{\beta_{G_j}\}_{j=1}^m$. $\{\lambda_j\}_{j=1}^m$ and $\{\tilde{\lambda}_i\}_{i=1}^p$ are two non-increasing positive sequences, and we defined the optimal sequences as follows, as well as in the analysis:

1. We define the sequence $\{\lambda_j\}_{j=1}^m$ as

$$\lambda_j \triangleq \sigma \sqrt{\log \frac{2ed}{s_0} + \frac{2}{s_0} \log \frac{4em}{j}}, \quad j \in [m], \quad (4)$$

where e denotes the base of the natural logarithm.

2. Then we extend this sequence to $\{\tilde{\lambda}_i\}_{i=1}^p$ as

$$\tilde{\lambda}_i \triangleq \begin{cases} \lambda_{\lceil \frac{i}{s_0} \rceil}, & i \leq s_0 \times m, \\ \lambda_m, & i > s_0 \times m. \end{cases} \quad (5)$$

Here we replicate each λ_j in $\{\lambda_j\}_{j=1}^m$ for s_0 times, and fill the remaining dimensions with λ_m to form a p -dimensional vector. And $\lceil \cdot \rceil$ is the ceiling function to get a real number's nearest integer neighbor from above.

Now we define the sparse group SLOPE penalty: Given a p -dimensional positive non-increasing sequence $\{\tilde{\lambda}_i\}_{i=1}^p$ in (5), we define

$$\|\beta\|_{\tilde{\lambda}^*} \triangleq \sum_{i=1}^p \tilde{\lambda}_i |\beta_{(i)}|. \quad (6)$$

Given a m -dimensional positive non-increasing sequence λ in (4), we define

$$\|\beta\|_{G, \lambda^*} \triangleq \sum_{j=1}^m \lambda_j \|\beta_{G(j)}\|_2, \quad (7)$$

Combined with the definitions in (4) - (7), we have the sparse group SLOPE penalty as follows:

$$\|\beta\|_* \triangleq \|\beta\|_{\tilde{\lambda}^*} + \sqrt{s_0} \|\beta\|_{G, \lambda^*}. \quad (8)$$

1.3. Related literature

The challenging problem of sparse group selection has garnered considerable attention over the past one to one and a half decades. In their respective works, [15] and [40] introduced a novel approach known as sparse group LASSO. This method simultaneously achieves element-wise and group sparsity by combining the LASSO penalty [42] with the group LASSO penalty [47], namely, $\ell_1 + \ell_{1,2}$ penalty. Recently, the minimax lower bound for double sparse linear regression was first derived by [9]. Furthermore, [9] obtained the sample complexity and the minimax upper bounds for sparse group LASSO, revealing that the method achieves the minimax optimality up to a logarithmic term. However, they state that the minimax optimality of sparse group LASSO cannot be achieved under the restricted isometry property (RIP [11, 10]) or restricted eigenvalue (RE) condition [5], and they use the irrepresentable condition [45] instead.

Whether the minimax optimality of sparse group LASSO can be achieved exclusively through RE or RIP conditions is a challenging and valuable problem. In the context of high-dimensional linear regression and compressed sensing, the RIP and RE conditions are fundamental assumptions. Indeed, such RE-type conditions are not merely sufficient but essentially necessary for the Lasso to achieve fast convergence rates. For instance, [12] provides a concrete example demonstrating that, in the absence of these conditions, the Lasso suffers from a slow rate of convergence.

For the analysis of LASSO, the irrepresentable condition [45] is a crucial requirement for ensuring support recovery. However, it is not a prerequisite for achieving optimal estimation error. In this paper, we establish the estimation error bound for sparse group LASSO based on RE-type and normalization conditions, which is entirely distinct from the technique used in [9]. Our approach has two main advantages:

- The RE and normalization conditions exhibit the differences and connections between single sparsity and double sparsity in a clearer way than the irrepresentable condition:
 - Our normalization condition (Assumption 1) explicitly incorporates the influence of group structure. Sparse group sparsity can thus be interpreted as a structural compromise between element-wise and group sparsity.
 - To establish the RE condition, it is essential to define a cone under both element-wise sparsity and group sparsity. The cone defined in Assumption 2 represents the intersection of these two distinct cones. This formulation elucidates that the double sparse parameter space is exactly the intersection of the individual sparse parameter spaces corresponding to element-wise and group sparsity.

These significant viewpoints cannot be revealed through the irrepresentable condition.

- The RE-type condition holds with high probability for certain classes of random matrices [36, 52, 22], as it depends solely on the matrix spectrum. In contrast, the irrepresentable condition lacks this verifiability. To fulfill our theoretical framework, we present the optimal sample complexity in the context of a random design.

In parallel with our work, several notable contributions have emerged [1, 49]. In particular, [1] shares a similar roadmap to ours. However, their technical assumptions—used to establish an upper bound on the empirical Gaussian complexity—appear challenging to verify and limit

the generalizability of their approach to other problems. In contrast, the conditions on which our approach relies can be readily satisfied under certain random design settings and yield optimal lower bounds on the sample size, which constitutes the core of our proof strategy. [49] studies the Double Sparse Iterative Hard Threshold (DSIHT) algorithm, and they prove the minimax optimality of DSIHT under the RIP-type condition. However, DSIHT requires much more stringent assumptions on the design matrix than the LASSO-type algorithm. Specifically, the IHT-type method requires an upper bound for the eigenvalues of the design matrix, while LASSO solely relies on the RE condition, which is more suitable for the highly correlated design matrix setting.

Table 1
Comparison of different works for double sparse linear regression

Method	Assumption	Theoretical Result	
		Support Recovery reliance free	Estimation Rate
SGLASSO [9]	Irrepresentable condition	×	$\frac{\sigma}{\sqrt{n}} \sqrt{s \log m + s s_0 \log \frac{ed}{s_0}}$
DSIHT [49]	DSRIP	×	$\frac{\sigma}{\sqrt{n}} \sqrt{s \log \frac{em}{s} + s s_0 \log \frac{ed}{s_0}}$
Ours	RE-type condition + Normalization	✓	$\frac{\sigma}{\sqrt{n}} \sqrt{s \log \frac{em}{s} + s s_0 \log \frac{ed}{s_0}}$

As illustrated in Table 1, we compare our work with [9] on sparse group LASSO and [49] on DSIHT. And we claim that our assumptions are the weakest. To summarize our main results, firstly, we only require two mild matrix conditions: the Sparse Group Normalization condition (Assumption 1) and the Restricted Eigenvalue condition (Assumptions 2 and 3). Secondly, we demonstrate that sparse group LASSO can achieve sharp optimality, improving the results of [9].

On the other hand, the proof of the estimation error for LASSO can be based on support recovery and primal-dual witness technique, such as [45], or on techniques from Gaussian complexity [4]. In a certain sense, [9] and ours respectively extend these two ideas under the framework of the double sparse structure. Our assumptions are the weakest since our technique does not rely on the results of support recovery and ℓ_∞ risk.

Inspired by the idea of our proof, we also construct the non-increasing weighted convex penalty $\|\beta\|_{\tilde{\lambda}^*} + \|\beta\|_{G, \lambda^*}$, which is also known as the sparse group SLOPE [14]. Before that, SLOPE [6] and Group SLOPE [8] are inspired by the minimax optimal properties of the Benjamini-Hochberg procedure [2] in the context of multiple testing. Our idea is mainly based on the inequality technique employed in the proof of the main results. Naturally, we can establish the minimax optimality of sparse group SLOPE under a new type of RE condition, which is different from the setting under an orthogonal design matrix by [41], yet similar to [4].

1.4. Our contributions

The main contributions of our work can be summarized as follows:

- We present a novel theoretical approach to derive the Gaussian complexity under the double sparse parameter space. Our idea eliminates the technical barriers between double sparsity and individual sparsity. This analysis reveals that the combination of normalization and restricted eigenvalue condition is sufficient for deriving the sharp optimal rate of sparse group LASSO.
- By leveraging this technique, we establish an estimation upper bound corresponding to sparse group SLOPE. This method is different from previous analyses about SLOPE, i.e. [6, 8], and can be extended to the more complicated sparse structure. We also establish the minimax lower bounds for double sparse regression. In the proof of lower bounds, we only need a normalization assumption, which is more general than previous works [9, 27].
- We further extend our theory to the random design setting. Specifically, we establish sufficient sample complexities to ensure the fulfillment of the normalization condition (Assumption 1) and the restricted eigenvalue conditions (Assumptions 2 and 3). This analysis is conducted under two distinct settings: the classical sub-Gaussian design [52] and the weaker moment framework adopted by [22].

1.5. Organization of the paper

In Section 2, we provide the main techniques to control the Gaussian complexity under the double sparse structure. Mainly building upon the results from Section 2.2, we derive the estimation error bounds for sparse group LASSO and sparse group SLOPE in Section 3 and Section 4, respectively. In Section 5, we establish the lower bound for ℓ_2 estimation error. Next, we extend our analysis to the case of random design in Section 6. A numerical study is presented in Section 7. Lastly, we discuss and interpret our results in Section 8.

2. Complexity analysis under double sparsity

2.1. Notation and preliminaries

For any positive integer p , we denote the set $\{1, 2, \dots, p\}$ as $[p]$. Denote $I(\cdot)$ as the indicator function. For any $a, b \in \mathbb{R}$, we use $a \vee b = \max(a, b)$ and $a \wedge b = \min(a, b)$. The floor function $\lfloor a \rfloor$ is the largest integer no greater than a , and the ceiling function $\lceil a \rceil$ is the smallest integer no less than a . We use the notation $a \lesssim b$ to denote the existence of a constant C independent of n such that $a \leq Cb$ holds uniformly, and correspondingly, we use $a \gtrsim b$. If $a \lesssim b$ and $a \gtrsim b$ hold simultaneously, we denote $a \asymp b$. Denote I_n as the $n \times n$ identity matrix. Given an n -dimensional vector u , we define $\|u\|_n^2 \triangleq \frac{1}{n} \sum_{i=1}^n u_i^2$. Given index set S , $u_S \in \mathbb{R}^{|S|}$ represents the subvector of u indexed by set S , and $X_S \in \mathbb{R}^{n \times |S|}$ represents the submatrix of X with columns indexed by S . For a matrix U , $\|U\|_{op}$ represents the spectral norm of U . We use $\|\cdot\|_2$ to represent the Frobenius norm of a matrix or the ℓ_2 -norm of a vector. For a vector u , define its corresponding matrix form $U \triangleq \text{mat}(u)$ such that $u = \text{vec}(U)$, where $\text{vec}(\cdot)$ is the vectorization operator.

To analyze element-wise sparsity and group sparsity simultaneously, we use the notation “ $(\cdot)$ ” in subscript to denote sorting by either the absolute values of elements or the ℓ_2 norm of vectors.

Definition 2.1 (Three types of rearrangement). Given a p -dimensional vector u , which can be rearranged into a $d \times m$ matrix U by m groups such that $U = \text{mat}(u)$, we define

- **Element-wise sorting:** Let $(u_{(i)})_{i=1}^p$ denote the sorted vector of $u \in \mathbb{R}^p$, such that $|u_{(1)}| \geq |u_{(2)}| \geq \dots \geq |u_{(p)}|$, where $|u_{(i)}|$ denotes the i -th largest absolute value of u .
- **Group-wise sorting:** We order the columns $U_j, j \in [m]$ of the matrix U by ℓ_2 norm, such that $\|U_{(1)}\|_2 \geq \|U_{(2)}\|_2 \geq \dots \geq \|U_{(m)}\|_2$, where $\|U_{(j)}\|_2$ denotes the j -th largest in ℓ_2 -norm among the m columns.
- **Within Group sorting:** Given the j -th group of u , i.e., the j -th column of U , $|U_{(i),j}|$ denotes the i -th largest absolute value in the j -th column, such that $|U_{(1),j}| \geq \dots \geq |U_{(d),j}|$.

2.2. Bound on the stochastic errors

In this section, we present a sparse group normalization condition for the design matrix X , which is crucial for our study of the sparse group penalty.

Assumption 1 (Sparse Group Normalization). *For each X_{G_j} , given any subset $S \subseteq G_j$ with $|S| = s_0$, the submatrix $X_S \in \mathbb{R}^{n \times s_0}$ satisfies $\frac{1}{\sqrt{n}} \|X_S\|_{op} \leq 1$, that is,*

$$\sup_{j \in [m]} \sup_{|S|=s_0, S \subseteq G_j} \|X_S\|_{op} \leq \sqrt{n}.$$

According to Assumption 1, if $\xi_i \sim N(0, \sigma^2)$ for any $i \in [n]$, and given any subset $S \subseteq G_j$ with $|S| = s_0$, there exists a random variable $Z \sim \chi_{s_0}^2$ (the χ^2 random variable with s_0 degrees of freedom) such that for any given $t > 0$, we have

$$\mathbf{P} \left(\frac{1}{n\sigma^2} \|\xi^\top X_S\|_2^2 \geq t \right) \leq \mathbf{P}(Z \geq t). \quad (9)$$

Remark 1. *The above assumption is the double-sparse analogue of the group normalization assumption in [30]. In specific, they impose $\|X_{G_j}\|_{op} \leq \sqrt{n}$ and then apply a χ^2 tail bound to $\|X_{G_j}^\top \xi\|_2^2$ for $j \in [m]$, which is the key step for the theoretical analysis under group sparsity. In our setting, each important group contains on average s_0 nonzero elements, so we need the same control for any $S \subseteq G_j$ with $|S| = s_0$ for any $j \in [m]$, and therefore assume that $\frac{1}{n} \|X_S\|_{op}^2$ is bounded by an absolute constant. When $s_0 = d$ this reduces to the condition in [30]; when $s_0 = d = 1$ it reduces to the usual normalization in [4, 37].*

By the definition of $\{\lambda_j\}_{j=1}^m$ and $\{\tilde{\lambda}_i\}_{i=1}^p$ in (4) and (5), we further define the positive homogeneous function $N(u) : \mathbb{R}^p \rightarrow [0, \infty)$ as follows:

$$N(u) \triangleq \frac{1}{\sqrt{n}} \left\{ \sum_{j=1}^m \|U_{(j)}\|_2 \sqrt{s_0} \lambda_j + \sum_{i=1}^p |u_{(i)}| \tilde{\lambda}_i \right\}, \quad (10)$$

where $u \in \mathbb{R}^p$ and $U \in \mathbb{R}^{d \times m}$ is the matrix form of u .

We now present the main concentration inequality for convex optimization under the double sparse structure:

Theorem 2.2. For any $\Delta_0 \in (0, 1)$, assume that the design matrix X satisfies Assumption 1 and $\xi \sim N(0, \sigma^2 I_n)$. Then, we have

$$\mathbf{P} \left\{ \forall u \in \mathbb{R}^p : \frac{1}{n} \xi^\top Xu \leq (4 + \sqrt{2}) \max \left(N(u), \|Xu\|_n \sigma \sqrt{\frac{\log(1/\Delta_0)}{n}} \right) \right\} \geq 1 - \frac{\Delta_0}{2}. \quad (11)$$

Theorem 2.2 is the key point of proving the error bounds of sparse group LASSO and SLOPE. Specifically speaking, combining with Assumption 1 for design matrix X , we can construct an envelope function $N(u)$ (10) to control the supremum of the Gaussian process. We give a proof sketch and the main idea of Theorem 2.2 in Appendix B.

Remark 2. We note that concurrent work by [26, 1] has studied sparse group LASSO and SLOPE in contexts of multi-task regression and multi-class classification. While Theorem 3 in [26] shares similarities with our Theorem 2.2, it is important to emphasize that our result is established under weaker assumptions. This technical improvement is crucial for handling double sparsity structures and forms a cornerstone of our theoretical framework.

3. Main results for sparse group LASSO

In this section, our goal is to investigate the statistical properties of the estimator $\hat{\beta}$ of the sparse group LASSO. We establish the estimation error bounds for $\hat{\beta}$ under our proposed RE-type conditions, which we call the strong sparse group restricted eigenvalue (SSGRE) condition.

Assumption 2 (Strong Sparse Group Restricted Eigenvalue Condition). Given design matrix $X \in \mathbb{R}^{n \times p}$, if X satisfies

$$\theta(s, s_0, c_0) \triangleq \min_{\delta \in C_{SSGRE}(s, s_0, c_0) \setminus \{0\}} \frac{\|X\delta\|_n}{\|\delta\|_2} > 0,$$

where $C_{SSGRE}(s, s_0, c_0) \triangleq \{\delta \in \mathbb{R}^p : \|\delta\|_1 + \sqrt{s_0} \|\delta\|_{1,2} \leq (2 + c_0) \sqrt{s s_0} \|\delta\|_2\}$ for some positive constant c_0 , we say X satisfies SSGRE condition with parameter $\theta(s, s_0, c_0)$.

Given $\gamma \in (0, 1)$, define

$$\varrho(\lambda) \triangleq \exp \left(- \left(\frac{\gamma \lambda \sqrt{n}}{(4 + \sqrt{2}) \sigma} \right)^2 \right) \quad \text{so that} \quad \lambda = \frac{(4 + \sqrt{2}) \sigma}{\gamma} \sqrt{\frac{\log(1/\varrho(\lambda))}{n}}.$$

We define the tuning parameter as $\lambda^\# = \frac{(4 + \sqrt{2}) \sigma}{\gamma \sqrt{n}} \sqrt{\log \frac{2ed}{s_0} + \frac{2}{s_0} \log \frac{4em}{s}}$. As discussed before, the penalty term of sparse group LASSO is defined as

$$h(\beta) \triangleq 2\lambda^\# \{ \|\beta\|_1 + \sqrt{s_0} \|\beta\|_{1,2} \}. \quad (12)$$

Theorem 3.1. Let $\Delta_0 \in (0, 1)$. Assume that $\beta^\star$ is (s, s_0) -sparse in model 1, and Assumption 1 holds. Assume that Assumption 2 holds with parameter $\theta(s, s_0, \frac{2(1+\gamma)}{1-\gamma})$. Then, with probability at least $1 - \frac{\Delta_0}{2}$, the solution of sparse group LASSO satisfies

$$\|\hat{\beta} - \beta^\star\|_2 \leq (1 + \gamma) \sqrt{s s_0} \lambda^\# \max \left\{ \frac{2}{\theta^2(s, s_0, \frac{2(1+\gamma)}{1-\gamma})}, \frac{1}{2} \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#)) s s_0} \right) \right\}.$$

Theorem 3.1 establishes the estimation upper bounds for sparse group LASSO based on the result (11) in Theorem 2.2.

Remark 3. In Theorem 3.1, let

$$\Delta_0 = \exp \left\{ -C_1 \left(ss_0 \log \frac{2ed}{s_0} + 2s \log \frac{4em}{s} \right) \right\}.$$

Then, $\left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#))ss_0} \right)$ is of constant order. Consequently, given a constant C_1 , there exists a sufficiently large constant C_2 such that, with probability at least $1 - \frac{\Delta_0}{2}$, the estimator $\hat{\beta}$ satisfies

$$\|\hat{\beta} - \beta^*\|_2 \leq C_2 \sigma \sqrt{\frac{1}{n} \left(ss_0 \log \frac{2ed}{s_0} + 2s \log \frac{4em}{s} \right)}. \quad (13)$$

4. Main results for sparse group SLOPE

This section provides the algorithmic construction and corresponding estimation properties of the sparse group SLOPE defined in the subsection 1.2.2, and we recall the weight sequences $\{\lambda_j\}_{j=1}^m$ and $\{\tilde{\lambda}_i\}_{i=1}^p$ defined by (4) and (5). Similar to the condition proposed in [4], we propose the Weighted Sparse Group Restricted Eigenvalue Condition (WSGRE):

Assumption 3 (Weighted Sparse Group Restricted Eigenvalue Condition). Given non-increasing weight sequences $\{\lambda_j\}_{j=1}^m$ and $\{\tilde{\lambda}_i\}_{i=1}^p$, the cone $C_{WSGRE}(s, s_0, c_0)$ is defined as

$$C_{WSGRE}(s, s_0, c_0) \triangleq \left\{ \delta \in \mathbb{R}^P : \|\delta\|_* \leq (2 + c_0) \sqrt{\sum_{i=1}^{ss_0} \tilde{\lambda}_i^2 \|\delta\|_2}, c_0 > 0 \right\}.$$

Given design matrix $X \in \mathbb{R}^{n \times P}$, if X satisfies

$$\theta(s, s_0, c_0) \triangleq \min_{\delta \in C_{WSGRE}(s, s_0, c_0) \setminus \{0\}} \frac{\|X\delta\|_n}{\|\delta\|_2} > 0, \quad (14)$$

we say that X satisfies the WSGRE condition with parameter $\theta(s, s_0, c_0)$.

The following argument demonstrates that for any $\delta \in C_{SSGRE}(s, s_0, c_0)$ (cf Assumption 2), it satisfies that $\delta \in C_{WSGRE}(s, s_0, 4 + c_0)$:

Based on the relationship between $\{\lambda_j\}$ and $\{\tilde{\lambda}_i\}$, we have $\tilde{\lambda}_{ss_0} = \lambda_s$ and $\sum_{i=1}^{ss_0} \tilde{\lambda}_i = s_0 \sum_{j=1}^s \lambda_j$. Assume that $\delta \in C_{SSGRE}(s, s_0, c_0)$, i.e., $\|\delta\|_1 + \sqrt{s_0} \|\delta\|_{1,2} \leq (2 + c_0) \sqrt{ss_0} \|\delta\|_2$. Observe that

$$\begin{aligned} \sqrt{s_0} \sum_{j=s+1}^m \lambda_j \|\delta_{G(j)}\|_2 + \sum_{i=s_0+1}^p \tilde{\lambda}_i |\delta_{(i)}| &\leq \lambda_s \left(\sum_{j=s+1}^m \sqrt{s_0} \|\delta_{G(j)}\|_2 + \sum_{i=s_0+1}^p |\delta_{(i)}| \right) \\ &\leq \lambda_s (\|\delta\|_1 + \sqrt{s_0} \|\delta\|_{1,2}) \\ &\leq (2 + c_0) \sqrt{ss_0} \lambda_s \|\delta\|_2. \end{aligned}$$

According to the Cauchy-Schwarz inequality, we have

$$\|\delta\|_* \leq \sqrt{s_0} \sum_{j=1}^s \lambda_j \|\delta_{G(j)}\|_2 + \sum_{i=1}^{ss_0} \tilde{\lambda}_i |\delta_{(i)}|$$

$$\begin{aligned}
& +\sqrt{s_0} \sum_{j=s+1}^m \lambda_j \|\delta_{G_{(j)}}\|_2 + \sum_{i=s s_0+1}^P \tilde{\lambda}_i |\delta_{(i)}| \\
& \leq \sqrt{s_0} \sqrt{\sum_{i=1}^s \lambda_i^2} \|\delta\|_2 + \sqrt{\sum_{i=1}^{s s_0} \tilde{\lambda}_i^2} \|\delta\|_2 + (2 + c_0) \sqrt{s s_0} \lambda_s \|\delta\|_2 \\
& \leq (6 + c_0) \sqrt{s s_0} \lambda_s \|\delta\|_2.
\end{aligned}$$

In other words, the WSGRE condition is more stringent than the SSGRE condition. Consider the penalty of sparse group SLOPE as

$$h(\beta) = \frac{2(4 + \sqrt{2})}{\sqrt{n}\gamma} \|\beta\|_*, \quad (15)$$

where γ is some constant belonging to $(0, 1)$. The following theorem establishes the estimation upper bound for the sparse group SLOPE.

Theorem 4.1. Assume that β^* is (s, s_0) -sparse in model 1, and Assumption 1 holds. Assume that Assumption 3 holds with parameter $\theta(s, s_0, \frac{2(1+\gamma)}{1-\gamma})$. If Δ_0 is the same as in Remark 3, then with probability at least $1 - \frac{\Delta_0}{2}$, the solution of sparse group SLOPE satisfies

$$\|\hat{\beta} - \beta^*\|_2 \leq (1 + \gamma) \sqrt{s s_0} \lambda^\# \max \left\{ \frac{2}{\theta(s, s_0, \frac{2(1+\gamma)}{1-\gamma})^2}, \frac{1}{2} \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#)) s s_0} \right) \right\}, \quad (16)$$

where $\lambda^\#$ and $\varrho(\lambda)$ are defined in the same manner as in Theorem 3.1.

The conclusion of Theorem 4.1 is similar to the conclusion of Theorem 3.1 for sparse group LASSO. However, the matrix condition in Theorem 4.1 is stronger compared to the condition used in Theorem 3.1. The advantage of sparse group SLOPE over sparse group LASSO is akin to the advantage of ordinary SLOPE over LASSO, where it becomes adaptive to the unknown sparsity s . The reason behind this “adaptivity to s ” is that the RE condition has been slightly strengthened for sparse group SLOPE.

5. Minimax lower bound under normalization assumption

In this section, we provide the minimax lower bounds for ℓ_2 -parameter estimation on the parameter space $\mathcal{DS}(s, s_0)$ and state that the upper bounds established in Theorem 3.1 and 4.1 match the lower bounds. Previous works have extensively studied the minimax rates for high-dimensional sparse linear regression. Numerous research papers have concentrated on the class of s -sparse elements, as exemplified by works such as [37, 44, 4]. Simultaneously, there have been endeavors dedicated to exploring group sparsity, as shown in [30].

Though a similar minimax lower bound of double sparse regression has been provided in [9, 49], We derive the same lower bound solely by relying on Assumption 1, without any RE or RIP conditions. We begin with the definition of the following parameter subspace. Consider the parameter space $\Theta^{m,d}(s, s_0)$:

$$\Theta^{m,d}(s, s_0) \triangleq \{\beta \in \mathbb{R}^P : \|\beta\|_{0,2} \leq s \text{ and } \|\beta_{G_j}\|_0 \leq s_0, \forall j \in [m]\}.$$

Different from $\mathcal{DS}(s, s_0)$, $\Theta^{m,d}(s, s_0)$ requires each group to be constrained to an ℓ_0 -ball with radius s_0 , and the group sparsity of $\Theta^{m,d}(s, s_0)$ reaches s . It is obvious that $\Theta^{m,d}(s, s_0) \subseteq \mathcal{DS}(s, s_0)$. Hence we have

$$\inf_{\hat{\beta}} \sup_{\beta^* \in \mathcal{DS}(s, s_0)} \mathbb{E} \|\hat{\beta} - \beta^*\|_2^2 \geq \inf_{\hat{\beta}} \sup_{\beta^* \in \Theta^{m,d}(s, s_0)} \mathbb{E} \|\hat{\beta} - \beta^*\|_2^2. \quad (17)$$

In what follows, we consider the minimax lower bounds for the parameter space $\mathcal{DS}(s, s_0)$. Let $M(\rho; \Theta^{m,d}(s, s_0), \rho_H)$ be the cardinality of ρ -packing set of the parameter space $\Theta^{m,d}(s, s_0)$ in the sense of Hamming metric ρ_H . We first provide the lower bounds for the packing number of $\Theta^{m,d}(s, s_0)$.

Lemma 5.1 (Lower bounds for the packing number [27]). *The cardinality of $\frac{ss_0}{4}$ -packing set of $\Theta^{m,d}(s, s_0)$ with hamming distance is lower bounded as*

$$M\left(\frac{ss_0}{4}; \Theta^{m,d}(s, s_0), \rho_H\right) \geq \exp \left\{ \frac{1}{4} \left(s \log \frac{em}{s} + ss_0 \log \frac{ed}{s_0} \right) \right\}.$$

[27] utilized the structures of double sparsity and combined multi-ary Gilbert-Varshamov bounds [16] to construct the packing set of $\Theta^{m,d}(s, s_0)$ in a more concise manner.

Existing proofs of the minimax lower bound for double sparse structures [9, 27] rely on an eigenvalue condition for the design matrix. In contrast, to establish the same lower bound without this assumption, we instead construct a signed packing set via the following lemma, which is built on Lemma 5.1 and Assumption 1. Denote

$$\vartheta_{\max} = \sup_{j \in [m]} \sup_{|S|=s_0, S \subseteq G_j} \|X_S\|_{op} / \sqrt{n}. \quad (18)$$

Lemma 5.2. *Assume that there exists a constant $\vartheta_{\max} > 0$ such that the design matrix $X \in \mathbb{R}^{n \times p}$ satisfies (18). Then, there exists a subset $\mathcal{M} \subseteq \Theta^{m,d}(s, s_0) \cap \{-1, 0, 1\}^p$ satisfying the following properties:*

- (i) Given $\beta \in \mathcal{M}$, $\|\beta\|_{0,2} = s$ and $\|\beta_{G_j}\|_0 = s_0, \forall j \in [m]$.
- (ii) $\|X\beta\|_n^2 \leq \vartheta_{\max}^2 ss_0, \forall \beta \in \mathcal{M}$.
- (iii) Given $\beta^i, \beta^j \in \mathcal{M}$, the hamming distance satisfies $\rho_H(\beta^i, \beta^j) \geq \frac{ss_0}{4}$.

Combining Lemma 5.2, we establish the minimax lower bound in the following theorem.

Theorem 5.3. *Consider linear regression model $y = X\beta^* + \xi$, where $\xi \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$. Assume that X satisfies (18) with $\vartheta_{\max} < \infty$. Then, we have*

$$\inf_{\hat{\beta}} \sup_{\beta^* \in \mathcal{DS}(s, s_0)} \mathbb{E} \|\hat{\beta} - \beta^*\|_2^2 \geq \frac{\sigma^2}{128n\vartheta_{\max}^2} \left(s \log \frac{em}{s} + ss_0 \log \frac{ed}{s_0} \right). \quad (19)$$

Theorem 5.3 establishes the lower bounds for the ℓ_2 estimation errors, which are consistent with the results in [9, 49]. Notably, drawing inspiration from the work of [4], we establish the lower bound by exclusively leveraging the sparse group normalization condition (18).

6. Random design

This section explores conclusions under random designs and is divided into two parts: the first part focuses on weak random design that is studied by [22], where there is no need for a specific distribution assumption, such as sub-Gaussian or any specific distribution. The second part is for the traditional sub-Gaussian random design. It is different from [52] in that we do not need RIP-type conditions but only the normalization assumption 1 combined with restricted eigenvalue assumption 2 or 3. We mainly establish random design properties for sparse group SLOPE, and the conclusions for sparse group LASSO hold similarly. To summarize, we clarify that Assumptions 1, 2, and 3 can hold with high probability in the setting of random design.

6.1. Sparse group structure from the geometry viewpoint

The restricted eigenvalue assumption 2 for sparse group LASSO is based on strong RE defined by [4], where they define the cone for the element-wise parameter space $\mathcal{S}_0(ss_0) \triangleq \{\beta \in \mathbb{R}^p : \|\beta\|_0 \leq ss_0\}$:

$$C_{SRE}(ss_0, c_1) \triangleq \{\delta \in \mathbb{R}^p : \|\delta\|_1 \leq (1 + c_1)\sqrt{ss_0}\|\delta\|_2, c_1 > 0\}.$$

LASSO can obtain the optimal estimation rate by RE condition on $C_{SRE}(ss_0, c_1)$, i.e., ss_0 -sparsity. On the other hand, we can define another cone for group sparsity $\mathcal{S}_{0,2}(s) \triangleq \{\beta \in \mathbb{R}^p : \|\beta\|_{0,2} \leq s\}$:

$$C_{SGRE}(s, c_2) \triangleq \{\delta \in \mathbb{R}^p : \|\delta\|_{1,2} \leq (1 + c_2)\sqrt{s}\|\delta\|_2, c_2 > 0\}.$$

Considering the parameter space $\mathcal{S}_0(ss_0) \cap \mathcal{S}_{0,2}(s)$, the cone $C_{SSGRE}(s, s_0, c_0)$ plays an important role in Assumption 2. Let $c_0 = c_1 + c_2$, we have

$$C_{SRE}(ss_0, c_1) \cap C_{SGRE}(s, c_2) \subseteq C_{SSGRE}(s, s_0, c_0).$$

In addition, we have

$$C_{SSGRE}(s, s_0, c_0) \subseteq C_{SRE}(ss_0, 1 + c_0) \cap C_{SGRE}(s, 1 + c_0).$$

Therefore, neglecting the constant factor, we can approximate the cone C_{SSGRE} using the form $C_{SRE} \cap C_{SGRE}$. This approximation is analogous to the fact that $\mathcal{DS}(s, s_0) = \mathcal{S}_0(ss_0) \cap \mathcal{S}_{0,2}(s)$.

On the other hand, [4, 22] demonstrate a delicate conclusion that the RE condition and the sparse eigenvalue are equivalent, and we can trivially obtain the equivalence between the group RE condition and the group sparse eigenvalue condition. As a result, we can make a conjecture that there exists a double sparse version equivalence between Assumption 2 and the double sparse eigenvalue condition.

6.2. Weak random design for sparse group LASSO

Without loss of generality, in this section, we assume the columns of the design matrix X are normalized by the factor $\frac{1}{\sqrt{n}}$, and we give a formal version of the double sparse eigenvalue condition:

Assumption 4. If the design matrix X satisfies

$$\theta \triangleq \min_{\beta \in \mathcal{DS}(s, s_0)} \frac{\|X\beta\|_2}{\|\beta\|_2} > 0,$$

then X is said to satisfy the double sparse eigenvalue condition $\text{DSRE}(s, s_0, \theta)$.

We conclude with the following theorem to demonstrate the equivalence between the double sparse eigenvalue condition and the strong sparse group RE condition. The technique here is based on Maurey's empirical method, which has been used in [22, 33].

Theorem 6.1. Assume that X satisfies Assumption 1 and Assumption 4. Then, there exists a $\tilde{s}$ of the same order as s , and a constant $\tilde{\theta}$ of the same order as θ that make X satisfy the condition $\text{SSGRE}(\tilde{s}, s_0, c_0, \tilde{\theta})$.

Theorem 6.1 implies that the combination of Assumptions 1 and 4 can deduce Assumption 2. Specifically, under an i.i.d. random design, if we verify the corresponding conditions that satisfy Assumptions 1 and 4, then Assumption 2 naturally holds as a consequence of Theorem 6.1.

A useful condition for the sparse eigenvalue condition is the small ball condition [21, 22, 31], and [4] follows from it. Next, we show the double sparse version of the small ball condition.

Theorem 6.2 (Small ball condition). A random vector $x \in \mathbb{R}^P$ satisfies the small ball condition over $\mathcal{DS}(s, s_0)$ if there exist positive constants τ and $\theta_{\min}$ such that

$$\mathbf{P}(|\langle \beta, x \rangle| \geq \theta_{\min} \|\beta\|_2) \geq \tau, \quad \forall \beta \in \mathcal{DS}(s, s_0). \quad (20)$$

Let X be a random design matrix whose rows are i.i.d. copies of x satisfying (20). If the sample size satisfies

$$n \geq \frac{C}{\tau^2} \left(s \log \frac{em}{s} + s s_0 \log \frac{ed}{s_0} \right),$$

then, with probability at least $1 - \exp(-C'n\tau^2)$, the matrix X satisfies the $\text{DSRE}(s, s_0, \theta)$ condition for some constant $\theta \geq \theta_{\min}/\sqrt{2}$, where $C, C' > 0$ are absolute constants.

Remark 4. Theorem 6.2 extends Corollary 2.5 in [22] to the double sparsity. This generalization requires bounding the VC dimension for the double sparse structure, which replaces the corresponding term for ordinary sparsity in the original result.

To extend Assumption 1 to the setting of random design, we also follow the insight in [22] and derive a double sparse version. We define the family of the index set

$$\mathcal{S} \triangleq \{S : |S| = s_0 \text{ and } \exists i, S \subseteq G_i\}. \quad (21)$$

Based on (21), we define s_0 -dimensional random vector $X_j, j \in \mathcal{S}$. We define X_j^* as the $\frac{1}{\sqrt{s_0}}$ -scaled dual form of $X_j, j \in \mathcal{S}$, i.e.,

$$X_j^* \triangleq \sup_{\|t\|_2 \leq \frac{1}{\sqrt{s_0}}} |\langle X_j, t \rangle| = \frac{1}{\sqrt{s_0}} \|X_j\|_2, \quad \forall j \in \mathcal{S}.$$

In other words, X_j^* is the operator norm of $\frac{1}{\sqrt{s_0}} X_j$ induced by $\|\cdot\|_2$. Assumption 1 is exactly the upper bound of $\max_{j \in \mathcal{S}} \frac{1}{n} \sum_{i=1}^n (X_{ij}^*)^2$ with high probability, where X_{ij}^* is the i -th copy of X_j^* .

Theorem 6.3 (Weak moment condition). *Given random variable Z , $\|Z\|_{L_q}$ denotes $\{\mathbf{E}(|Z|^q)\}^{\frac{1}{q}}$. Let $q_0 = C_1(\log m + s_0 \log \frac{ed}{s_0})$, where $C_1 > 1$ is an absolute constant. Assume that for any $j \in \mathcal{S}$, $\|X_j^*\|_{L_2} \leq 1$, and $\forall 2 \leq q \leq q_0$, $\|X_j^*\|_{L_q}$ satisfies the growth rate $\|X_j^*\|_{L_q} \leq \kappa_1 q^\alpha$ for $\alpha \geq \frac{1}{2}$. Assume that the sample size $n \geq C_2(\alpha) \kappa_1^2 q_0^{4\alpha-1}$, where $C_2(\alpha) = 4\alpha \exp(4\alpha - 1)$ is a constant depending on α and larger than 1. Then, Assumption 1 holds with probability at least $1 - \exp\{-(C_2 - 1)(\log m + s_0 \log \frac{ed}{s_0})\}$.*

If we consider the sub-Gaussian random design, we have the growth rate $\|X_j^*\|_{L_q} \leq \kappa_1 q^{\frac{1}{2}}$ for any $q \geq 2$. So the condition in Theorem 6.3 is weaker because we only need the growth rate to hold for $q \leq q_0$. q_0 exists because we use a truncation technique in the proof. A tight sufficient sample size n is another key point for random design. When $\alpha = \frac{1}{2}$, we have $n \asymp q_0$, which exactly meets the sample size lower bound we need for sub-Gaussian random design. Therefore, Theorem 6.3 delivers an exact relationship between moment growth rate and the sample size, which helps us tackle different random design settings, such as sub-Gaussian for $\alpha = \frac{1}{2}$ and sub-exponential for $\alpha = 1$.

Therefore, combining Theorem 6.2 and Theorem 6.3, we achieve assumptions 1 and 2 with probability at least $1 - \exp\{-C_1(s \log \frac{em}{s} + ss_0 \log \frac{ed}{s_0})\} - \exp\{-C_2(\log m + s_0 \log \frac{ed}{s_0})\}$. Moreover, if we set $\alpha = \frac{1}{2}$ in Theorem 6.3, where sub-Gaussian design is one of the special cases, we derive a tight sample size rate, i.e., $n \geq C_3(s \log \frac{em}{s} + ss_0 \log \frac{ed}{s_0}) + C_4(\log m + s_0 \log \frac{ed}{s_0})$.

6.3. Sub-Gaussian random design

In this section, we will discuss the sample complexity needed to satisfy the sparse group normalization assumption 1 and the restricted eigenvalue assumptions 2 and 3 under the assumption of sub-Gaussian random design. Moreover, we derive the results based on the covariance matrix condition. To do this, we need to introduce some definitions, which can be found in [52, 32].

Definition 6.4 (Isotropic sub-Gaussian [38]). Let Y be a random vector in $\mathbb{R}^p$

1. Y is called isotropic if for every $y \in \mathbb{R}^p$, $\mathbf{E}|\langle Y, y \rangle|^2 = \|y\|_2^2$.
2. Y is ψ_2 with a constant α if for every $y \in \mathbb{R}^p$,

$$\|\langle Y, y \rangle\|_{\psi_2} \triangleq \inf \left\{ t : \mathbf{E} \exp \left(\langle Y, y \rangle^2 / t^2 \right) \leq 2 \right\} \leq \alpha \|y\|_2.$$

The ψ_2 condition on a scalar random variable V is equivalent to the subgaussian tail decay of V , which means for some constant c ,

$$\mathbf{P}(|V| > t) \leq 2 \exp \left(-t^2 / c^2 \right), \text{ for all } t > 0.$$

An important example of an isotropic, sub-Gaussian random vector is the Gaussian random vector $Y = (h_1, h_2, \dots, h_p)$ where $h_i, \forall i \in [p]$ are independent $\mathcal{N}(0, 1)$ random variables. Another example is the Bernoulli random vector $Y = (\epsilon_1, \dots, \epsilon_p)$, where $\epsilon_i, \forall i \in [p]$ are independent, symmetric ± 1 Bernoulli random variables.

Suppose $\zeta_1, \zeta_2, \dots, \zeta_n$ are independent and identically distributed p -dimensional isotropic, sub-Gaussian random vectors, forming a random matrix $Z \in \mathbb{R}^{n \times p}$, whose i -th row is denoted by ζ_i . In this paper, we consider a random design matrix X , which is generated as follows:

$$X \triangleq Z\Sigma^{\frac{1}{2}}, \quad (22)$$

where Σ is the covariance matrix. Here, we need to consider an appropriate Σ so that the empirical design matrix X can satisfy the RE and normalization assumptions with high probability. Specifically, according to the theoretical framework of [52, 32], given the vector space $\mathcal{V} \in \mathbb{R}^p$, the key point is to construct the restricted isometric properties between Xv and $\Sigma^{\frac{1}{2}}v$, and then the condition for empirical design matrix X can be transformed into the corresponding condition for Σ . To achieve this goal, the technique from empirical process plays an important role, and we define Gaussian complexity first:

Definition 6.5. Given a subset $\mathcal{V} \in \mathbb{R}^p$, we define the Gaussian complexity of $\mathcal{V}$ as follows:

$$\ell^*(\mathcal{V}) \triangleq \mathbf{E} \sup_{\delta \in \mathcal{V}} \left| \sum_{i=1}^p g_i \delta_i \right|,$$

where δ_i is each component of vector δ , and $g_1, g_2, \dots, g_p$ are independent $\mathcal{N}(0, 1)$ random variables. In particular, given a non-negative definite matrix Σ , we define:

$$\tilde{\ell}^*(\mathcal{V}) \triangleq \ell^*(\Sigma^{\frac{1}{2}}\mathcal{V}) = \mathbf{E} \sup_{v \in \mathcal{V}} \left| \langle \Sigma^{\frac{1}{2}}v, g \rangle \right| = \mathbf{E} \sup_{v \in \mathcal{V}} \left| \langle v, \Sigma^{\frac{1}{2}}g \rangle \right|.$$

According to the homogeneity of the norm, we only need to consider the subset of the unit ball sphere S^{p-1} , which is defined as:

$$S^{p-1} \triangleq \{v \in \mathbb{R}^p : \|v\|_2 = 1\}.$$

The main technique we use is the following empirical process result:

Lemma 6.6 (Theorem 2.1 in [32]). *Let $1 \leq n \leq p$ and $0 < \theta < 1$. Let ζ be an isotropic sub-Gaussian random vector on $\mathbb{R}^p$, and constant ψ_2 be α . And $\zeta_1, \dots, \zeta_n$ are independent copies of ζ . Let X be the random matrix defined in (22), and let $\{\Sigma^{\frac{1}{2}}v, v \in \mathcal{V}\} \subset S^{p-1}$. If the sample size n satisfies $n > c'\alpha^4\theta^2\tilde{\ell}^*(\mathcal{V})^2$. Then, with probability at least $1 - \exp(-\bar{c}\theta^2n/\alpha^4)$, for all $v \in \mathcal{V}$, we have*

$$1 - \theta \leq \frac{\|Xv\|_2}{\sqrt{n}} \leq 1 + \theta,$$

where $c', \bar{c} > 0$ are some absolute constants.

Therefore, given some specific parameter space $\mathcal{V}$, we will set Σ to guarantee that for any $v \in \mathcal{V}$, $\Sigma^{\frac{1}{2}}v \in S^{p-1}$ (in fact, we just need $0 < \|\Sigma^{\frac{1}{2}}v\|_2 < \infty$), and we derive the Gaussian complexity $\tilde{\ell}^*(\mathcal{V})$.

6.3.1. Sparse group normalization in sub-Gaussian random design

Consider the parameter space

$$V_{s_0} \triangleq \{v : \text{supp}(v) \in \mathcal{S}\} \cap \{v : \Sigma^{\frac{1}{2}}v \in S^{p-1}\}, \quad (23)$$

where $\text{supp}(v)$ represents the support set of v and $\mathcal{S}$ is given in (21). Similarly, we define

$$\tilde{V}_{s_0} \triangleq \{v : \text{supp}(v) \in \mathcal{S}\} \cap \{v : \|\Sigma^{\frac{1}{2}}v\|_2 \leq 1\}.$$

It is easy to observe that $V_{s_0} \subseteq \tilde{V}_{s_0}$.

Theorem 6.7. For $1 < s_0 < \frac{d}{2}$, it holds that

$$\ell^*(V_{s_0}) \leq 6 \left(\log m + s_0 \log \frac{5ed}{s_0} \right)^{\frac{1}{2}}.$$

So that for any set $S \in \mathcal{S}$, assume that the sub-matrix Σ_S satisfies that $\lambda_{\max}(\Sigma_S) \leq \frac{1}{(1+\theta)^2}$. If sample size n satisfies that for some constant $0 < \theta < 1$,

$$n > \frac{c'\alpha^4}{\theta^2} \left(\log m + s_0 \log \frac{ed}{s_0} \right), \quad (24)$$

then with probability at least $1 - \exp(-\bar{c}\theta^2 n/\alpha^4)$, we have for all $v \in \{v : \text{supp}(v) \in \mathcal{S}\}$,

$$\frac{\|Xv\|_n}{\|v\|_2} \leq 1,$$

where $c', \bar{c} > 0$ are absolute constants.

6.3.2. SSGRE condition and WSGRE condition under sub-Gaussian random design

In this section, we derive the sub-Gaussian random design version of two RE conditions we used for sparse group LASSO and SLOPE.

As mentioned above, the calculation of Gaussian complexity is the key point. Owing to the technique of Proposition B.4, we can derive the complexity for these two estimators. We demonstrate that the condition we require is weaker than the classic sub-Gaussian random design result [52].

Because we have given the random design property for sparse group normalization, in this part, we assume that sparse group normalization holds. We have the following Theorem to ensure the validity of weighted sparse group RE. Strong sparse group RE holds as follows.

Theorem 6.8 (WSGRE for sub-Gaussian random design). Let the covariance matrix Σ satisfy the RE condition on $C_{WSGRE}(s, s_0, c_0)$, that is,

$$\min_{v \in C_{WSGRE}(s, s_0, c_0) \setminus \{0\}} \frac{\|\Sigma^{\frac{1}{2}}v\|_2}{\|v\|_2} = \kappa \in (0, 1), \quad (25)$$

and the sparse group normalization condition, that is,

$$\sup_{S \in \mathcal{S}} \|\Sigma_S\|_2 \leq 1, \text{ for all } S \in \mathcal{S}.$$

Let the generation of random design matrix $X = Z\Sigma^{\frac{1}{2}} \in \mathbb{R}^{n \times p}$, if the sample size n satisfies

$$n > C_1 \left(s \log \frac{4em}{s} + ss_0 \log \frac{2ed}{s_0} \right),$$

then with the probability not less than $1 - \exp(-C_2 n)$, $\forall v \in C_{WSGRE}(s, s_0, c_0) \setminus \{0\}$, we have

$$\frac{\|Xv\|_2}{\|v\|_2} \geq \kappa',$$

where $C_1, C_2 > 0$ are absolute constants and $\kappa' \in (0, \kappa)$.

Corollary 1 (SSGRE for sub-Gaussian random design). *If the covariance matrix satisfies the SSGRE condition, that is,*

$$\min_{v \in C_{SSGRE}(s, s_0, c_0) \setminus \{0\}} \frac{\|\Sigma^{\frac{1}{2}} v\|_2}{\|v\|_2} = \kappa \in (0, 1).$$

Assume all the conditions in Theorem 6.8 hold. If the sample size n satisfies

$$n > C_1 \left(s \log \frac{4em}{s} + s s_0 \log \frac{2ed}{s_0} \right),$$

then with the probability greater than $1 - \exp(-C_2 n)$, $\forall v \in C_{SSGRE}(s, s_0, c_0) \setminus \{0\}$, we have

$$\frac{\|Xv\|_2}{\|v\|_2} \geq \kappa',$$

where $C_1, C_2 > 0$ are absolute constants and $\kappa' \in (0, \kappa)$.

Proof. Because $C_{SSGRE}(s, s_0, c_0) \subseteq C_{WSGRE}(s, s_0, 2 + c_0)$, so we can immediately get the Gaussian complexity on $C_{SSGRE}(s, s_0, c_0) \cap \{u : \|\Sigma^{\frac{1}{2}} u\|_2 = 1\}$. The rest of the proof is the same as Theorem 6.8. $\square$

Remark 5. The approach of [52] needs

$$\rho_{\max} = \sup_{\beta \in \mathcal{DS}(s, s_0)} \frac{\|\Sigma_S\|_2}{\|\beta\|_2} < \infty.$$

Since each parameter β in $\mathcal{DS}(s, s_0)$ uniquely determines the support set, and the support of $\mathcal{DS}(s, s_0)$ contains $\mathcal{S}$ (as defined in (21), $\mathcal{S}$ is equal to the support of $\mathcal{DS}(1, s_0)$), this assumption is strictly stronger than the sparse group normalization used in Theorem 6.8. This difference comes from the technique of [4]. Our results are obtained under normalization and RE conditions only, without irrerepresentable conditions [9] or RIP-type assumptions.

7. Numerical experiments

In this section, we carry out a series of simulation studies to verify our theoretical findings. Specifically, we compare sparse group LASSO (SGLASSO) with both LASSO and Group LASSO (GLASSO), and sparse group SLOPE (SGSLOPE) with both SLOPE and Group SLOPE (GSLOPE). These comparisons aim to demonstrate the advantages of SGLASSO and SGSLOPE over the individual selection method in parameter estimation and variable selection within the double sparse linear model.

We generate the simulation data using a linear model represented as $y = X\beta^* + \xi$, where $\xi \sim N(0, \sigma^2 I_n)$. We consider two simulation settings:

- **Scenario 1:** $m = 100, d = 20, s = 5, s_0 = 10, \sigma = 1$. n increases from 100 to 1000 in increments of 100.

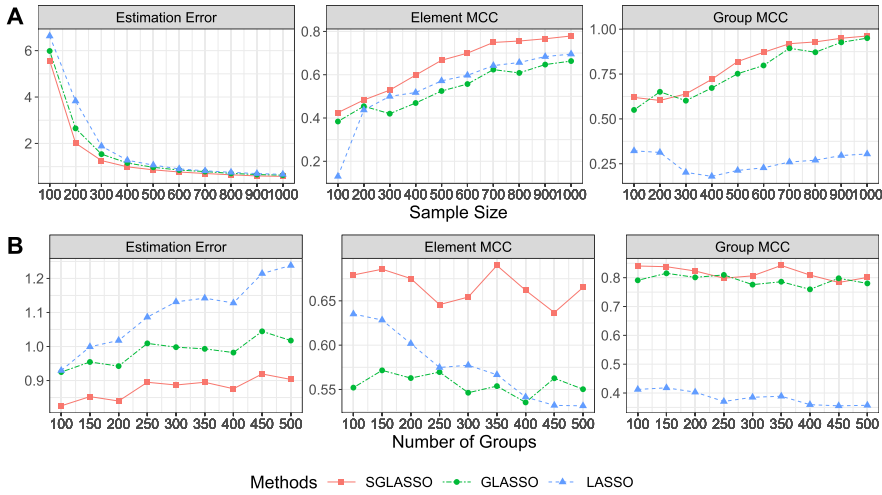


Fig 1. Performance on different metrics for the LASSO-type methods as the sample size or the number of groups increases. (A) Computational results for SGLASSO, GLASSO, and LASSO under scenario 1. (B) Computational results for SGLASSO, GLASSO, and LASSO under scenario 2.

- **Scenario 2:** $n = 300, d = 10, s = 5, s_0 = 5, \sigma = 1$. m increases from 100 to 500 in increments of 50.

For each setting, we randomly draw $X \in \mathbb{R}^{n \times md}$ and the non-zero components of β^* with i.i.d. standard normal entries.

For the tuning parameters, we use a grid search technique as [9] showed to tune the unknown parameters λ and s_0 . Here, we use 5-fold cross-validation to tune the model parameters. Referring to [9], we set the candidate values of s_0 as a sequence $\{s_{0,l}\}_{l=1}^L = \{d^{\frac{l-1}{L-1}}, 1 \leq l \leq L\}$, where we set $L = 5$ in our numerical experiment. Here we implement our methods based on R packages `sparsegl` [28] and `sgs` [14].

All the simulation results are based on 50 repetitions. Given an output $(\hat{S}, \hat{\beta})$, we assess the performance of each method by examining the estimation error, denoted as $\|\hat{\beta} - \beta^*\|_2$. Moreover, we use Mathew's Correlation Coefficient (MCC) to assess the performance of model selection, which is defined as

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}},$$

where $\text{TP} = \hat{S} \cap S^*$ and $\text{TN} = \hat{S}^c \cap (S^*)^c$ stand for true positives/negatives, respectively. $\text{FP} = \hat{S} \cap (S^*)^c$ and $\text{FN} = \hat{S}^c \cap S^*$ stand for false positives/negatives, respectively. Here, we consider MCC at both the element and group levels. The computational results are depicted in Figure 1 and 2.

In Figure 1, all methods demonstrate strong performance as n increases. SGLASSO has better performances than GLASSO and LASSO on all the metrics, and LASSO has a significant drawback in group selection than the other two methods. As n becomes larger, the differences diminish, and all methods exhibit commendable performance in estimating β^* . On the other hand, when m increases, the estimation error of all methods exhibits a significant increasing

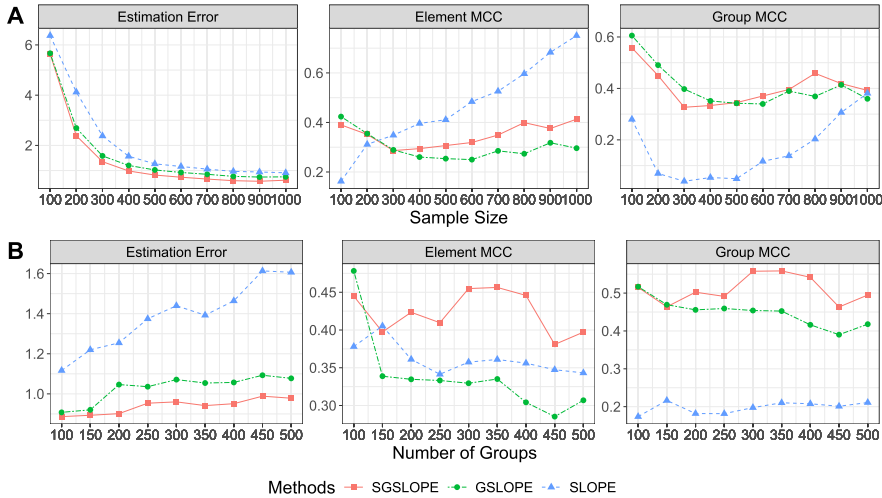


Fig 2. Performance on estimation error for the SLOPE-type methods as the sample size or the number of groups increases. (A) Computational results for SGSLOPE, GSLOPE, and SLOPE under scenario 1. (B) Computational results for SGSLOPE, GSLOPE, and SLOPE under scenario 2.

trend. Moreover, SGLASSO outperforms the other two methods in model selection at both element and group levels.

Figure 2 presents a similar performance as the LASSO-type methods shown in Figure 1. SGSLOPE outperforms or has a comparative performance with both GSLOPE and SLOPE.

8. Discussion and future work

In this paper, we study the high-dimensional double sparse regression, in which both element-wise and group-wise sparsity exist in the regression coefficient. From the perspective of the convex algorithm, we study two kinds of $\ell_1 + \ell_{1,2}$ algorithms: the universal tuning parameter for sparse group LASSO and the weighted tuning parameter for sparse group SLOPE. We derive the estimation upper bounds for both and clarify that they achieve sharp optimality in the minimax sense.

Several important problems can be addressed in the future. Firstly, an effective algorithm for sparse group SLOPE can be developed, building upon the ideas proposed in our current work. Secondly, the issue of adaptation to the unknown sparsity s_0 remains to be solved. One potential approach to achieve adaptation to s_0 is through some minimax adaptation procedure, which has been widely used and studied in the related field of statistics [25, 24, 13, 3]. Moreover, our approach presents a novel solution to address simultaneous sparse structures, making it a versatile and valuable technique applicable to various domains and applications. For instance, it can be applied to simultaneously sparse and low-rank structure recovery [34, 17] as well as sparse tensor SVD [48].

A. Additional definition and notation

To simplify the notations, denote the length p ($= d \times m$) vector

$$\phi = \frac{1}{\sqrt{n}} \xi^\top X$$

where ξ and X are random noise and design matrix in (1) and

$$\Phi = \text{mat}(\phi) \in \mathbb{R}^{d \times m}$$

as its matrix form, with columns indexed by the groups. And we have $\phi = \text{vec}(\Phi)$. With a little abuse of notation, we use Φ_S as its vector version ϕ_S here or there, where S is a subset of indices in $[p]$. To further analyze the double sparse structure, we consider two types of index sets as follows:

- Firstly, select s groups from m groups at random. Then, choose $s \times s_0$ elements from the selected groups to form a subvector with dimensions $s \times s_0$. The collection of index sets corresponding to these subvectors is referred to as the collection of sets $\mathbb{S}_1(s, s_0)$.
- Firstly, select s groups from m groups at random. Then, choose s_0 elements from each selected group, resulting in a subvector with dimensions $s \times s_0$. The index sets corresponding to the subvectors obtained in this manner form a collection of sets, denoted by $\mathbb{S}_2(s, s_0)$. Obviously, $\mathbb{S}_2(s, s_0) \subsetneq \mathbb{S}_1(s, s_0)$.

B. Proof for Theorem 2.2

To prove Theorem 2.2, we first provide some useful lemmas. Assume that s_0 , d , and m are known fixed constants in the following discussion. For a given $s \in [m]$, we consider two forms of subvectors, which can be denoted as

$$\Psi_1(s) \triangleq \sup_{S \in \mathbb{S}_1(s, s_0)} \frac{1}{ss_0\sigma^2} \|\phi_S\|_2^2$$

and

$$\Psi_2(s) \triangleq \sup_{S \in \mathbb{S}_2(s, s_0)} \frac{1}{ss_0\sigma^2} \|\phi_S\|_2^2.$$

Next, we derive the upper bound of $\Psi_1(s)$ and $\Psi_2(s)$ based on Assumption 1.

Lemma B.1. Assume that Assumption 1 holds. Then, we have

$$\mathbf{P} \left\{ \Psi_1(s) \geq \frac{16}{3} \left(\log \frac{2ed}{s_0} + \frac{2}{s_0} \log \frac{4em}{s} \right) \right\} \leq \frac{s}{4m}, \quad (26)$$

and

$$\mathbf{P} \left\{ \Psi_2(s) \geq \frac{8}{3} \left(\log \frac{2ed}{s_0} + \frac{2}{s_0} \log \frac{4em}{s} \right) \right\} \leq \frac{s}{4m}. \quad (27)$$

Now we define two random variables ν_s and Υ_s , and present the upper bounds for the random variables Υ_s and ν_s , which hold uniformly over $s \in [m]$.

Definition B.2 (Definition of ν_s and Υ_s). We define two random variables ν_s and Υ_s which are corresponding to $\mathbb{S}_1(s, s_0)$ and $\mathbb{S}_2(s, s_0)$, respectively:

- By the rule of $\mathbb{S}_1(s, s_0)$, we obtain ss_0 elements each time, and we only focus on the ss_0 -th largest one. ν_s denotes the ss_0 -th largest element of Φ_S for any $S \in \mathbb{S}_1(s, s_0)$. Consequently, we have $\nu_s^2 \leq \Psi_1(s)$.

- By the rule of $\mathbb{S}_2(s, s_0)$, we obtain s groups each time, and we only focus on the s -th largest group with ℓ_2 -norm. Υ_s denotes the ℓ_2 -norm of the s -th largest group of Φ_S for any $S \in \mathbb{S}_2(s, s_0)$, so that we have $\Upsilon_s^2 \leq s_0 \Psi_2(s)$.

Lemma B.3. Assume that Assumption 1 holds. Define the event

$$\Omega \triangleq \left\{ \max_{j \in [m]} \frac{\Upsilon_j}{\lambda_j} \leq 4\sqrt{s_0} \right\} \cap \left\{ \max_{j \in [m]} \frac{v_j}{\lambda_j} \leq 4 \right\}.$$

Then, we have

$$\mathbf{P}(\Omega) \geq \frac{1}{2}.$$

Lemma B.3 demonstrates that the maximum values of random variables Υ_s and v_s can be upper-bounded by λ_j with high probability.

Recall that we define the sequence $\{\lambda_j\}_{j=1}^m$ as

$$\lambda_j \triangleq \sigma \sqrt{\log \frac{2ed}{s_0} + \frac{2}{s_0} \log \frac{4em}{j}}, \quad j \in [m],$$

the non-increasing sequence $\{\tilde{\lambda}_i\}_{i=1}^p$:

$$\tilde{\lambda}_i \triangleq \begin{cases} \lambda_{\lceil \frac{i}{s_0} \rceil}, & i \leq s_0 \times m, \\ \lambda_m, & i > s_0 \times m. \end{cases} \quad (28)$$

and recall the positive homogeneous function $N(u) : \mathbb{R}^p \rightarrow [0, \infty)$ as follows:

$$N(u) = \frac{1}{\sqrt{n}} \left\{ \sum_{j=1}^m \|U_{(j)}\|_2 \sqrt{s_0} \lambda_j + \sum_{i=1}^p |u_{(i)}| \tilde{\lambda}_i \right\},$$

where $U = \text{mat}(u)$ and we have the following proposition:

Proposition B.4. Under Assumption 1 and ξ is defined in Theorem 2.2, with probability at least $\frac{1}{2}$, we have

$$\sup_{u \in \mathbb{R}^p : N(u) \leq 1} \left| \frac{1}{n} \xi^\top Xu \right| \leq 4.$$

Remark 6. If we derive the Proposition B.4, we can easily obtain Theorem 2.2 by the similar idea of [4]. However, the main difference is to capture the double sparse structure. To achieve this, we need Lemma B.1 and Lemma B.3 to control the complexity of double sparsity, and we need to emphasize that just normalization assumption (Assumption 1) is necessary.

Proof of Lemma B.1

Proof. Let $\Lambda \subseteq [m]$ be the s -length column index sets of ϕ_S such that the supremum in $\Psi_2(s)$ achieves. We consider upper bounding the moment generating function of $\Psi_2(s)$. For simplicity, let $\sigma^2 = 1$. For $t > 0$, we have

$$\mathbf{E} \{ \exp [t \cdot \Psi_2(s)] \} = \mathbf{E} \left\{ \exp \left[\frac{t}{s s_0} \sum_{j=1}^s \sum_{i=1}^{s_0} (\Phi_{i, \Lambda_j}^*)^2 \right] \right\}$$

$$\begin{aligned} &\leq \frac{1}{s} \sum_{j=1}^s \mathbf{E} \left\{ \exp \left[\frac{t}{s_0} \sum_{i=1}^{s_0} (\Phi_{i,\Lambda_j}^*)^2 \right] \right\} \\ &\leq \frac{1}{s} \sum_{j=1}^m \mathbf{E} \left\{ \exp \left[\frac{t}{s_0} \sum_{i=1}^{s_0} (\Phi_{(i),j})^2 \right] \right\}, \end{aligned} \quad (29)$$

where the first inequality follows from the AM-GM inequality.

Next, we prove the upper bounds of $\mathbf{E} \left\{ \exp \left(\frac{t}{s_0} \sum_{i=1}^{s_0} (\Phi_{(i),j})^2 \right) \right\}$. Take the j -th column as an example. It is easy to observe that

$$\begin{aligned} \exp \left(\frac{t}{s_0} \sum_{i=1}^{s_0} (\Phi_{(i),j})^2 \right) &\leq \sum_{\substack{S \subseteq [d] \\ |S|=s_0}} \exp \left(\frac{t}{s_0} \sum_{i \in S} \Phi_{i,j}^2 \right) \\ &\leq \binom{d}{s_0} \exp \left(\frac{t}{s_0} \sum_{i \in S} \Phi_{i,j}^2 \right). \end{aligned}$$

In the last inequality, the set S can be chosen as any eligible set, since under expectation the result is uniformly bounded by a χ^2 distribution. Taking the expectation on both sides, we can obtain

$$\mathbf{E} \left\{ \exp \left(\frac{t}{s_0} \sum_{i=1}^{s_0} (\Phi_{(i),j})^2 \right) \right\} \leq \binom{d}{s_0} \mathbf{E} \left\{ \exp \left(\frac{t}{s_0} \sum_{i=1}^{s_0} \Phi_{i,j}^2 \right) \right\}. \quad (30)$$

Let $Z \sim \chi_{s_0}^2$. Then the moment-generating function of Z is given by

$$M_Z(t) = (1 - 2t)^{-\frac{s_0}{2}}, \quad t < \frac{1}{2}.$$

Therefore, according to the Assumption 1, we have

$$\begin{aligned} \mathbf{E} \left\{ \exp \left[\frac{t}{s_0} \sum_{i=1}^{s_0} \Phi_{i,j}^2 \right] \right\} &\leq \mathbf{E} \left\{ \exp \left[\frac{t}{s_0} Z \right] \right\} \\ &= \left(1 - \frac{2t}{s_0} \right)^{-\frac{s_0}{2}}. \end{aligned} \quad (31)$$

Combining (29)-(31), according to Chernoff bound, we have

$$\begin{aligned} \mathbf{P}(\Psi_2(s) \geq a) &\leq \inf_{t \geq 0} \frac{\mathbf{E} \{ \exp(t \cdot \Psi_2(s)) \}}{e^{at}} \\ &\leq \frac{m}{s} \binom{d}{s_0} 2^{s_0} \exp\left(-\frac{3}{8} a s_0\right) \\ &\leq \frac{4m}{s} \exp \left\{ -\frac{3}{8} s_0 \left(a - \frac{8}{3} \log \frac{2ed}{s_0} \right) \right\}, \end{aligned}$$

where the second inequality follows from $t = \frac{3}{8} s_0$. Let $a = \frac{8}{3} (\log \frac{2ed}{s_0} + \frac{2}{s_0} \log \frac{4m}{s})$, we obtain

$$\mathbf{P} \left[\sup_{S \in \mathbb{S}_2(s, s_0)} \frac{1}{s s_0 \sigma^2} \|\phi_S\|_2^2 \geq \frac{8}{3} \left(\log \frac{2ed}{s_0} + \frac{2}{s_0} \log \frac{4m}{s} \right) \right] \leq \frac{s}{4m}.$$

Next, we turn to analyze the random variable $\Psi_1(s)$ on $\mathbb{S}_1(s, s_0)$. By the previous definition, $\Psi_1(s) \geq \Psi_2(s)$ holds almost surely. However, we prove that the tail probability inequalities of the two variables only differ by a constant factor.

Here we use the set family $\mathbb{S}_2(1, s_0)$ to “cover” any index set in $\mathbb{S}_1(s, s_0)$ and $\mathbb{S}_2(s, s_0)$, and the cardinality of $\mathbb{S}_2(1, s_0)$ is $|\mathbb{S}_2(1, s_0)| = m \binom{d}{s_0}$. For any index set $S \in \mathbb{S}_2(s, s_0)$, it can be separated into s subsets S^j , $j \in [s]$ by column, and $|S^j| = s_0$. Therefore, we have $\mathbb{S}_2(1, s_0)$ such that $S^j \in \mathbb{S}_2(1, s_0)$, $j \in [s]$, which we call S can be covered by $\mathbb{S}_2(1, s_0)$.

However, it is not sufficient to use s elements of $\mathbb{S}_2(1, s_0)$ to “cover” any index set $S \in \mathbb{S}_1(s, s_0)$. In what follows, we show that $2s$ sets of $\mathbb{S}_2(1, s_0)$ are sufficient to cover any $S \in \mathbb{S}_1(s, s_0)$. For any $S \in \mathbb{S}_1(s, s_0)$, let a_i be the number of elements in the i -th column of S , where we only need to consider s non-zero columns. Observe that

$$a_1 + a_2 + \cdots + a_s = ss_0.$$

For S^j , $j \in [s]$, we can use at most $\lfloor \frac{a_i}{s_0} \rfloor + 1$ sets of $\mathbb{S}_2(1, s_0)$ to cover S^j . Then according to the properties of the floor function, that is, $\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor$, we obtain

$$\sum_{i=1}^s \left\{ \left\lfloor \frac{a_i}{s_0} \right\rfloor + 1 \right\} \leq s + \left\lfloor \frac{\sum_{i=1}^s a_i}{s_0} \right\rfloor = 2s.$$

Therefore, we can cover S by $2s$ disjoint sets of set family $\mathbb{S}_2(1, s_0)$. Here we denote $\{F^j\}_{j=1}^{2s}$ as the covering set of $S \in \mathbb{S}_1(s, s_0)$. Similar to (29), we have

$$\begin{aligned} \mathbf{E} \{ \exp(t \cdot \Psi_1(s)) \} &= \mathbf{E} \left\{ \exp \left(\frac{1}{2s} \sum_{j=1}^{2s} \frac{2t}{s_0} \|\Phi_{F^j}\|_2^2 \right) \right\} \\ &\leq \frac{1}{2s} \sum_{j=1}^{2s} \mathbf{E} \left\{ \exp \left(\frac{2t}{s_0} \|\Phi_{F^j}\|_2^2 \right) \right\}. \end{aligned}$$

The rest of the proof is similar to the case of $\mathbb{S}_2(s, s_0)$. We obtain

$$\mathbf{P} \left[\sup_{S \in \mathbb{S}_1(s, s_0)} \frac{1}{ss_0\sigma^2} \|\Phi_S\|_2^2 \geq \frac{16}{3} \left(\log \frac{2ed}{s_0} + \frac{2}{s_0} \log \frac{4m}{s} \right) \right] \leq \frac{s}{4m}.$$

These complete the proofs of Lemma B.1. $\square$

Similar bounds have been obtained by [49]. However, [49] derived upper bounds under a more stringent condition, namely, the double sparse RIP condition, which results in a tighter probability tail bound. In our paper, we choose to relax this assumption and solely utilize the sparse group normalization condition.

Proof of Lemma B.3

Proof. Following the definitions, we can easily obtain that

$$\Upsilon_s \leq \sqrt{s_0 \Psi_2(s)} \quad \text{and} \quad \nu_s \leq \sqrt{\Psi_1(s)}.$$

Without loss of generality, we ignore the difference of constant factor and rewrite the conclusion of $\mathbb{S}_2(s, s_0)$ in Lemma B.1 as:

$$\mathbf{P} \left[\sup_{S \in \mathbb{S}_2(s, s_0)} \frac{1}{s s_0 \sigma^2} \|\phi_S\|_2^2 \geq \frac{16}{3} \left(\log \frac{2ed}{s_0} + \frac{2}{s_0} \log \frac{4m}{s} \right) \right] \leq \frac{s}{4m}.$$

The following is proved using the peeling technique [43]. With respect to Υ_s , according to (27), for any given $j \in [m]$, we have

$$\mathbf{P} \left(\Upsilon_j \leq \frac{4}{\sqrt{3}} \sqrt{s_0} \lambda_j \right) \geq 1 - \frac{s}{4m}, \quad j \in [m]. \quad (32)$$

Let q be the integer such that $2^q \leq m \leq 2^{q+1}$. Apply (32) to $j = 2^l$ for $l = 0, 1, 2, \dots, q-1$ and define the event Ω_1 defined as

$$\Omega_1 \triangleq \left\{ \max_{l=0,1,\dots,q-1} \frac{\Upsilon_{2^l} \sqrt{3}}{4\sqrt{s_0} \lambda_{2^l}} \leq 1 \right\}.$$

By the union bound, we have $P(\Omega_1) \geq 1 - \frac{\sum_{l=0}^{q-1} 2^l}{4m} \geq 1 - \frac{2^q - 1}{4m} \geq \frac{3}{4}$. For any $j < 2^q$, there exists $l \in \{0, 1, \dots, q-1\}$ such that $2^l < j < 2^{l+1}$, and, thus, with the satisfaction of event Ω_1 ,

$$\begin{aligned} \Upsilon_j &\leq \Upsilon_{2^l} \leq \frac{4\sigma}{\sqrt{3}} \sqrt{s_0 \log \frac{2ed}{s_0} + 2 \log \frac{4m}{2^l}} \\ &< 4\sigma \sqrt{\frac{s_0}{3} \log \frac{2ed}{s_0} + \frac{2}{3} \log \frac{8m}{j}} \\ &< 4\sigma \sqrt{s_0 \log \frac{2ed}{s_0} + 2 \log \frac{2m}{j}} \\ &< 4\sqrt{s_0} \lambda_j. \end{aligned}$$

On the other hand, for $2^q < j \leq m \leq 2^{q+1}$, there are

$$\begin{aligned} \Upsilon_j &\leq \Upsilon_{2^{q-1}} \leq \frac{4\sigma}{\sqrt{3}} \sqrt{s_0 \log \frac{2ed}{s_0} + 2 \log \frac{4m}{2^{q-1}}} \\ &< 4\sigma \sqrt{\frac{s_0}{3} \log \frac{2ed}{s_0} + \frac{2}{3} \log \frac{16m}{2^{q+1}}} \\ &< 4\sigma \sqrt{\frac{s_0}{3} \log \frac{2ed}{s_0} + \frac{2}{3} \log \frac{16m}{j}} \\ &< 4\sigma \sqrt{s_0 \log \frac{2ed}{s_0} + 2 \log \frac{4m}{j}} \\ &= 4\sqrt{s_0} \lambda_j. \end{aligned}$$

Thus, when event Ω_1 holds, we have $\Upsilon_j \leq 4\sqrt{s_0} \lambda_j$ for all $j = 1, 2, 3, \dots, m$.

Next, we turn to the proof of ν_s . We first show the monotonicity of ν_s , i.e., $\nu_s \geq \nu_{s+1}$ for all $s \in [m-1]$. By the definition of ν_{s+1} , there exists $s+1$ columns such that the $s_0 \times (s+1)$ -th

largest absolute value among these columns is v_{s+1} . The index set of these $s + 1$ columns denotes G . We preserve the largest $s_0 \times (s + 1)$ elements in G , and denote the column that v_{s+1} belongs to as g . We separate our discussion into two cases:

1. If column g contains fewer than s_0 elements, then the columns $G \setminus g$ contain more than ss_0 elements. Therefore, among these s columns, the ss_0 -th largest element is larger than v_{s+1} . By definition, it is smaller than v_s . Consequently, we can conclude that $v_s \geq v_{s+1}$.
2. If column g has at least s_0 elements, there exists column $g' \neq g$, which has no more than s_0 elements. Therefore, among $G \setminus g'$, the ss_0 -th largest element is larger than v_{s+1} . By definition, we have $v_s \geq v_{s+1}$.

The rest of the work is just a matter of combining (26) with the monotonicity of v_s . Similarly, we can obtain the event

$$\Omega_2 \triangleq \left\{ \max_{l=0,1,\dots,q-1} \frac{v_{2^l} \sqrt{3}}{4\lambda_{2^l}} \leq 1 \right\}$$

satisfying $P(\Omega_2) \geq \frac{3}{4}$. By the union bound, we have

$$P(\Omega_1 \cap \Omega_2) \geq \frac{3}{4} + \frac{3}{4} - 1 = \frac{1}{2}.$$

This completes the proof of Lemma B.3. $\square$

Proof of Proposition B.4

Before presenting the proof of Proposition B.4, we introduce the rearrangement inequality, a crucial concept frequently employed in the subsequent proofs.

Lemma B.5 (Rearrangement inequality [18]). *The rearrangement inequality states that for any selection of real numbers:*

$$x_1 \leq \dots \leq x_n \quad \text{and} \quad y_1 \leq \dots \leq y_n,$$

and for any permutation $x_{\sigma(1)}, \dots, x_{\sigma(n)}$ of $x_1, \dots, x_n$, it holds that

$$x_n y_1 + \dots + x_1 y_n \leq x_{\sigma(1)} y_1 + \dots + x_{\sigma(n)} y_n \leq x_1 y_1 + \dots + x_n y_n.$$

Proof. We define the matrix U^* as follows: U^* is obtained by first “within Group sorting” the matrix U and then taking the absolute value of each entry, such that $U_{i,j}^* = |U_{(i),j}|$, since we did not permute the columns of the matrix, we have $\|U_j^*\|_2 = \|U_j\|_2$. We make some illustration in the figure 3.

We separate the proof of Proposition B.4 into four steps.

Step 1(Decomposition into parts A_1 and A_2): According to the definition of matrices U, Φ defined in 2.1, we have

$$\left| \frac{1}{n} \xi^\top X u \right| = \left| \frac{1}{\sqrt{n}} \sum_{j=1}^m \sum_{i=1}^d U_{i,j} \Phi_{i,j} \right|.$$

S_1	S_2	S_3	S_4				
S_1	S_2	S_3	S_4				
S_1	S_3		S_4				
S_2							

Fig 3. Illustration of Φ^* . We let $m = 8$, $d = 7$ and $s_0 = 3$. The pink region represents A_1 , while the region below the pink corresponds to A_2 .

To establish the upper bound for the A_2 part, we further partition A_2 into several subsets such as S_1 , S_2 , S_3 and S_4 according to the division rule in Step 3 of the proof to Proposition B.4: for any two indices $j_1 < j_2$, we have $\Phi_{s_0+1,j_1}^* \geq \Phi_{s_0+1,j_2}^*$, then S_1 consists of s_0 largest elements in the first column, S_2 consists of s_0 largest elements in the first two columns except S_1 , and so on.

Through the rearrangement inequality for each column, we have

$$\left| \frac{1}{\sqrt{n}} \sum_{j=1}^m \sum_{i=1}^d U_{i,j} \Phi_{i,j} \right| \leq \frac{1}{\sqrt{n}} \sum_{j=1}^m \sum_{i=1}^d |U_{(i),j} \Phi_{(i),j}|.$$

Divide the above into two parts by rows, i.e., write

$$\frac{1}{\sqrt{n}} \sum_{j=1}^m \sum_{i=1}^d |U_{(i),j} \Phi_{(i),j}| = \underbrace{\frac{1}{\sqrt{n}} \sum_{j=1}^m \sum_{i=1}^{s_0} U_{i,j}^* \Phi_{i,j}^*}_{A_1} + \underbrace{\frac{1}{\sqrt{n}} \sum_{j=1}^m \sum_{i=s_0+1}^d U_{i,j}^* \Phi_{i,j}^*}_{A_2}.$$

Step 2(Upper bounds for part A_1):

For the part A_1 , by definition of U^* , Φ^* , we define the subvectors $U_{A_1,j}^*$, as

$$U_{A_1,j}^* = [U_{1,j}^*, U_{2,j}^*, \dots, U_{s_0,j}^*] = [|U_{(1),j}|, |U_{(2),j}|, \dots, |U_{(s_0),j}|],$$

as well as

$$\Phi_{A_1,j}^* = [\Phi_{1,j}^*, \Phi_{2,j}^*, \dots, \Phi_{s_0,j}^*] = [| \Phi_{(1),j} |, | \Phi_{(2),j} |, \dots, | \Phi_{(s_0),j} |].$$

Then, by the Cauchy-Schwarz inequality, we have

$$\begin{aligned} A_1 &\leq \frac{1}{\sqrt{n}} \sum_{j=1}^m \|U_{A_1,j}^*\|_2 \cdot \|\Phi_{A_1,j}^*\|_2 \\ &\leq \frac{1}{\sqrt{n}} \sum_{j=1}^m \|U_j^*\|_2 \cdot \|\Phi_{A_1,j}^*\|_2 \end{aligned}$$

$$\leq \frac{1}{\sqrt{n}} \sum_{j=1}^m \|U_{(j)}\|_2 \cdot \|\Phi_{A_1, (j)}^*\|_2,$$

where the last inequality follows the rearrangement inequality B.5.

Note that $\Phi_{A_1, (j)}^*$ is Υ_j as defined in Lemma B.3. Therefore, we have

$$\begin{aligned} A_1 &\leq \frac{1}{\sqrt{n}} \sum_{j=1}^m \|U_{(j)}\|_2 \cdot \Upsilon_j \\ &= \frac{1}{\sqrt{n}} \sum_{j=1}^m \|U_{(j)}\|_2 \cdot \sqrt{s_0} \lambda_j \frac{\Upsilon_j}{\sqrt{s_0} \lambda_j} \\ &\leq \frac{1}{\sqrt{n}} \sum_{j=1}^m \|U_{(j)}\|_2 \cdot \sqrt{s_0} \lambda_j \left(\max_{j \in [m]} \frac{\Upsilon_j}{\sqrt{s_0} \lambda_j} \right). \end{aligned} \quad (33)$$

Step 3 (Upper bounds for part A_2). For part A_2 , we first divide the index set of the part A_2 into some disjoint subsets $S_1, S_2, \dots$, where each subset is of size s_0 (here we assume that d is larger than $2s_0$, otherwise we can fill in with zeros). Then, we use the random variable v_s defined in Lemma B.3. As a result, we have

$$A_2 = \frac{1}{\sqrt{n}} \sum_q \sum_{(i,j) \in S_q} |U_{(i),j} \Phi_{(i),j}^*|.$$

Without loss of generality, we assume that the absolute values of the $s_0 + 1$ -th row of Φ^* are rearranged in descending order, such that for any $j_1 < j_2$, $\Phi_{s_0+1, j_1}^* \geq \Phi_{s_0+1, j_2}^*$. Next, we specify the construction of the sets $\{S_q\}$:

1. Select the largest s_0 elements from the first column of part A_2 to form set S_1 ;
2. Consider the first q ($q \geq 2$) columns of part A_2 , and select the largest s_0 elements from the remaining elements excluding set $\cup_{i=1}^{q-1} S_i$ to form set S_q .
3. Repeat step 2 $m - 1$ times to obtain $\{S_1, \dots, S_m\}$. Allocate the remaining elements arbitrarily into sets of size s_0 , and finally, obtain the family of sets $\{S_q\}$.

We also illustrate the $\{S_q\}$ by figure 3.

We denote the maximum value of Φ^* within set S_q by $\|\Phi_{S_q}^*\|_\infty$. Next, we prove that **for any given $q \in [m]$, there exists at least $s_0 \times q$ elements in the first q columns of Φ^* that are no less than $\|\Phi_{S_q}^*\|_\infty$.**

Remark 7. Φ^* only permutes the order of the elements in the same column of Φ , so it does not affect the definition of v_q . Therefore, if the claim above is clarified, we have $\|\Phi_{S_q}^*\|_\infty \leq v_q \leq 4\lambda_q$ for $q \in [m]$ when Lemma B.3 holds. For $q > m$, we have $\|\Phi_{S_m}^*\|_\infty \geq \|\Phi_{S_q}^*\|_\infty$ by the construction of $\{S_q\}$.

We prove the above claim using induction:

1. When $q = 1$, we have $\|\Phi_{S_1}^*\|_\infty = \Phi_{s_0+1, 1}^*$. Then we have

$$\Phi_{s_0+1, 1}^* \leq \Phi_{s_0, 1}^*,$$

which implies that the first s_0 elements of Φ_1^* are no less than $\|\Phi_{S_1}^*\|_\infty$. Therefore, we can conclude that there are s_0 elements no less than $\|\Phi_{S_q}^*\|_\infty$ in the first column.

2. For $q < m$, we assume that the claim holds. Note that there are at least $q \cdot s_0$ elements larger than $\|\Phi_{S_q}^*\|_\infty$ in the first q columns by assumption. For $q + 1$, we consider the following two cases:

- (a) $\|\Phi_{S_{q+1}}^*\|_\infty$ appears in the first q columns of part A_2 . By the construction of S_{q+1} , we have that all the elements of $\Phi_{S_q}^*$ are larger than $\|\Phi_{S_{q+1}}^*\|_\infty$. Since $|S_q| = s_0$, there are at least $q \cdot s_0 + s_0$ elements larger than $\|\Phi_{S_{q+1}}^*\|_\infty$ in the first q columns by induction.
- (b) $\|\Phi_{S_{q+1}}^*\|_\infty$ appears in the $(q+1)$ -th column. In this case, we exactly have $\|\Phi_{S_{q+1}}^*\|_\infty = \Phi_{s_0+1, q+1}^*$. Observe that

$$\Phi_{s_0+1, q+1}^* \leq \Phi_{s_0+1, q'}^* \leq \Phi_{s_0, q'}^*, \quad \forall q' \leq q + 1.$$

Thus there are at least $(q + 1) \cdot s_0$ elements larger than $\|\Phi_{S_{q+1}}^*\|_\infty$ in the first $q + 1$ columns.

Overall, we have proved that $\|\Phi_{S_q}^*\|_\infty \leq \nu_q$ for $q \in [m]$. For $q > m$, we obviously have $\|\Phi_{S_q}^*\|_\infty \leq \nu_m$. Arrange all elements of u in the A_2 part in descending order of absolute value, and the i -th position is noted as $|u_{A_2, (i)}|$.

$$\begin{aligned} A_2 &\leq \frac{1}{\sqrt{n}} \sum_q \sum_{(i, j) \in S_q} |U_{(i, j)}| \nu_{q \wedge m} \\ &\leq \frac{1}{\sqrt{n}} \sum_q \sum_{(i, j) \in S_q} |U_{(i, j)}| \lambda_{q \wedge m} \left(\max_{j \in [m]} \frac{\nu_j}{\lambda_j} \right) \\ &\leq \frac{1}{\sqrt{n}} \sum_{i=1}^{m \times (d-s_0)} |u_{A_2, (i)}| \tilde{\lambda}_i \left(\max_{j \in [m]} \frac{\nu_j}{\lambda_j} \right) \\ &\leq \frac{1}{\sqrt{n}} \sum_{i=1}^p |u_{(i)}| \tilde{\lambda}_i \left(\max_{j \in [m]} \frac{\nu_j}{\lambda_j} \right), \end{aligned} \tag{34}$$

where the third inequality follows from the rearrangement inequality.

Step 4 (Combination of part A_1 and A_2): When event Ω defined by Lemma B.3 satisfies that $\mathbf{P}(\Omega) \geq \frac{1}{2}$, (33) and (34) together show that

$$\begin{aligned} \sup_{u: N(u) \leq 1} \left| \frac{1}{n} \xi^\top Xu \right| &\leq A_1 + A_2 \\ &\leq \frac{1}{\sqrt{n}} \sum_{j=1}^m \|U_{(j)}\|_2 \sqrt{s_0} \lambda_j \left(\max_{j \in [m]} \frac{\Upsilon_j}{\sqrt{s_0} \lambda_j} \right) \\ &\quad + \frac{1}{\sqrt{n}} \sum_{i=1}^p |u_{(i)}| \tilde{\lambda}_i \left(\max_{j \in [m]} \frac{\nu_j}{\lambda_j} \right) \\ &\leq N(u) \cdot \left(\max_{j \in [m]} \frac{\Upsilon_j}{\sqrt{s_0} \lambda_j} \right) \vee \left(\max_{j \in [m]} \frac{\nu_j}{\lambda_j} \right) = 4. \end{aligned} \tag{35}$$

□

Proof of Theorem 2.2

The idea is similar to Proposition E.3 of [4] and can be obtained based on Proposition B.4, and we refine the proof here for the completeness of the article. The core idea of the proof of this theorem is the use of Levy-type inequalities for Lipschitz functions on Gaussian random variables, as specified in [7], Theorem 10.17 or [23], Eq. (1.4).

Proof. By homogeneity, it is enough to consider the set

$$T \triangleq \left\{ \forall u \in \mathbb{R}^p : \max \left(N(u), \frac{1}{L} \|Xu\|_n \right) \leq 1 \right\},$$

where $L \triangleq \left(\frac{n}{\sigma^2 \log(1/\Delta_0)} \right)^{\frac{1}{2}}$. Define function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$f(t) \triangleq \sup_{u \in T} \frac{1}{n} (\sigma t)^\top Xu \quad \text{for all } t \in \mathbb{R}^n.$$

It is obvious that $f(t)$ is a Lipschitz function with Lipschitz constant $\frac{\sigma L}{\sqrt{n}}$. Thus by [7], with probability at least $1 - \Delta_0/2$, we have

$$\begin{aligned} \sup_{u \in T} \frac{1}{n} \xi^\top Xu &\leq \text{Med} \left[\sup_{u \in T} \frac{1}{n} \xi^\top Xu \right] + \sigma L \sqrt{\frac{2 \log(1/\Delta_0)}{n}} \\ &\leq \text{Med} \left[\sup_{u: N(u) \leq 1} \frac{1}{n} \xi^\top Xu \right] + \sigma L \sqrt{\frac{2 \log(1/\Delta_0)}{n}} \\ &\leq 4 + \sigma L \sqrt{\frac{2 \log(1/\Delta_0)}{n}} = 4 + \sqrt{2}, \end{aligned}$$

where the last inequality uses $P(\Omega) \geq \frac{1}{2}$ to bound the median. $\square$

Proof sketch and insight for Theorem 2.2

In Theorem 2.2, we construct a function whose form is a weighted $\ell_1 + \ell_{1,2}$ norm, which we call the envelop function $N(u)$. By employing the techniques of the rearrangement inequality, we determine the weight of the ℓ_1 and $\ell_{1,2}$ norm and prove that $N(u)$ serves as an upper bound for the empirical Gaussian complexity $\frac{1}{n} \xi^\top Xu$.

One of the useful results is Proposition B.4 in Appendix B, which states that

$$\sup_{u \in \mathbb{R}^p : N(u) \leq 1} \left| \frac{1}{n} \xi^\top Xu \right| \leq 4$$

holds with probability at least $1/2$. This result and the Levy inequality (Theorem 10.17, [7]) directly leads to the result of Theorem 2.2.

The proof of Proposition B.4 relies on Lemma B.1 and Lemma B.3 in Appendix B. In the following part, we introduce the usage of these two lemmas:

1. In Lemma B.1, we derive the upper bound of the ℓ_2 -norm of the stochastic errors $\frac{1}{\sqrt{n}} \xi^\top X$ under double sparse settings, and the upper bound is of the order $\sigma \left(\log \frac{d}{s_0} + \frac{1}{s_0} \log \frac{m}{s} \right)^{1/2}$ with high probability. There is an innovative and important idea: Based on Assumption 1, we can leverage the moment-generating function of $\chi^2(s_0)$ to derive a Chernoff bound inequality for (9).

2. We define two key random variables ν_s and Υ_s in Definition B.2, and we can use the result in Lemma B.1 to upper bound these two terms. Lemma B.3 defines the event

$$\Omega \triangleq \left\{ \max_{j \in [m]} \frac{\Upsilon_j}{\lambda_j} \leq 4\sqrt{s_0} \right\} \cap \left\{ \max_{j \in [m]} \frac{\nu_j}{\lambda_j} \leq 4 \right\},$$

where λ_j is defined in (4). Then, based on the peeling technique and the results of Lemma B.1, Lemma B.3 shows that $\mathbf{P}(\Omega) \geq \frac{1}{2}$.

To leverage the results of Lemma B.1 and B.3, we need to employ some original inequality techniques. By [4], we are broadly aware that the Levy inequality can be useful. However, our work is far more delicate than the element-wise setting. Primarily, we will utilize the rearrangement inequality and induction. Specifically, We apply three sorting rules defined by Definition 2.1, and by ingeniously splitting and recombining the elements of the vector u , we derive the tightest rearrangement inequality and establish the upper bound for the Gaussian process $\sup_{u \in \mathbb{R}^P: N(u) \leq 1} \left| \frac{1}{n} \xi^\top Xu \right|$ (Proposition B.4):

$$\sup_{u: N(u) \leq 1} \left| \frac{1}{n} \xi^\top Xu \right| \leq \left(\max_{j \in [m]} \frac{\Upsilon_j}{\sqrt{s_0} \lambda_j} \right) \vee \left(\max_{j \in [m]} \frac{\nu_j}{\lambda_j} \right).$$

Combining Lemma B.3 and the above inequality, we prove Proposition B.4.

Proof for sub-Gaussian noise

Theorem 2.2 can be extended to sub-Gaussian noise for the same reason as Theorem 9.1 in [4] because we are only different in the form of the function $N(u)$. Assume that the noises $\xi = (\xi_1, \dots, \xi_n)$ are independent, have a zero mean, and exhibit sub-Gaussian behavior. Specifically, for some $\sigma > 0$,

$$\mathbf{E} \exp \left(\xi_i^2 / \sigma^2 \right) \leq 2, \quad i \in [n]. \quad (36)$$

By Proposition 9.2 in [4], denote $\mathcal{V}$ as

$$\mathcal{V} \triangleq \{u \in \mathbb{R}^P, \|Xu\|_n \leq 1, N(u) \leq 1\}.$$

For any $\Delta' > 0$, with probability at least $1 - \exp(-\Delta')$, we have

$$\begin{aligned} \sup_{u \in \mathcal{V}} \xi^\top Xu &\leq 8\mathbf{E} \left[\sup_{u \in \mathcal{V}} z^\top Xu \right] + 8\sqrt{2 \log \frac{1}{\Delta_0}} \\ &\leq 8\text{Med} \left[\sup_{u \in \mathcal{V}} z^\top Xu \right] + 8(\sqrt{\pi/2} + \sqrt{2\Delta'}), \end{aligned}$$

where z is a $\mathcal{N}(0, \sigma^2 \mathbf{I}_n)$ random vector and $\text{Med}[\cdot]$ denotes the median of a random variable. Note that we have $\text{Med}[\sup_{u \in \mathcal{V}} z^\top Xu] \leq 4$ in Proposition B.4. We have the following corollary:

Corollary 2. Assume that Assumption 1 holds and assume that ξ satisfies the sub-Gaussian assumption (36). Then, for a given $\Delta_0 \in (0, 1)$, with probability at least $1 - \frac{\Delta_0}{2}$, we have for all $u \in \mathbb{R}^P$:

$$\frac{1}{n} \xi^\top Xu \leq 40 \max \left(N(u), \sigma \|Xu\|_n \frac{\sqrt{\pi/2} + \sqrt{2 \log \left(\frac{1}{\Delta_0} \right)}}{\sqrt{n}} \right).$$

Generally speaking, $\frac{1}{n}\xi^\top Xu$ with Gaussian and sub-Gaussian noises share the same upper bounds up to some absolute constant. Therefore, the estimation rates can be extended to sub-Gaussian noise. For convenience, we analyze sparse group LASSO and SLOPE based on Theorem 2.2, and results for sub-Gaussian noises remain valid up to some absolute constants.

C. Proof of upper bounds for two estimators

Before formally stating our results, two useful lemmas are presented in the following part. Lemma C.1 is the basic inequality for double sparsity and Lemma C.2 can be applied to universal convex penalized linear regression problems. For these two Lemmas are similar to [4], so we omit the proof.

Lemma C.1. *Let $s \in [m]$ and $s_0 \in [d]$. For any two estimators $\beta, \hat{\beta} \in \mathbb{R}^p$, let $u = \hat{\beta} - \beta$. If $\|\beta\|_0 \leq ss_0$, we have*

$$\|\beta\|_{\tilde{\lambda}^*} - \|\hat{\beta}\|_{\tilde{\lambda}^*} \leq \left(\sum_{i=1}^{ss_0} \tilde{\lambda}_i^2 \right)^{\frac{1}{2}} \|u\|_2 - \sum_{i=ss_0+1}^p \tilde{\lambda}_i |u_{(i)}|. \quad (37)$$

If $\|\beta\|_{0,2} \leq s$, we have

$$\|\beta\|_{G,\lambda^*} - \|\hat{\beta}\|_{G,\lambda^*} \leq \left(\sum_{j=1}^s \lambda_j^2 \right)^{\frac{1}{2}} \|u\|_2 - \sum_{j=s+1}^m \lambda_j \|U_{(j)}\|_2. \quad (38)$$

Lemma C.2. *Let $h(\beta) : \mathbb{R}^p \rightarrow \mathbb{R}$ be a convex function. Consider model $y = X\beta + \xi$, and $\hat{\beta}$ be the solution to the convex optimization problem:*

$$\arg \min_{\beta \in \mathbb{R}^p} \|y - X\beta\|_n^2 + h(\beta). \quad (39)$$

Therefore, estimator $\hat{\beta}$ satisfies

$$\|X(\hat{\beta} - \beta^*)\|_n^2 \leq \frac{1}{n} \xi^\top X(\hat{\beta} - \beta^*) + \frac{1}{2} (h(\beta^*) - h(\hat{\beta})). \quad (40)$$

C.1. Proof of Theorem 3.1

Proof. According to the convex optimization problem (39) and Lemma C.2, we have

$$\|X(\hat{\beta} - \beta^*)\|_n^2 \leq \underbrace{\frac{1}{n} \xi^\top X(\hat{\beta} - \beta^*)}_{B_1} + \underbrace{\frac{1}{2} (h(\beta^*) - h(\hat{\beta}))}_{B_2}.$$

For part B_1 , recall the definition of $N(u)$ in (10). By the Cauchy-Schwarz inequality, we have

$$\begin{aligned} \sqrt{n}N(u) &\leq \left(\sum_{i=1}^{ss_0} \tilde{\lambda}_i^2 \right)^{\frac{1}{2}} \|u\|_2 + \sum_{i=ss_0+1}^p \tilde{\lambda}_i |u_{(i)}| + \left(\sum_{j=1}^s \lambda_j^2 \right)^{\frac{1}{2}} \sqrt{s_0} \|u\|_2 + \\ &\quad \sum_{j=s+1}^m \sqrt{s_0} \lambda_j \|U_{(j)}\|_2. \end{aligned} \quad (41)$$

Observe that

$$\begin{aligned} \left(\sum_{i=1}^{ss_0} \tilde{\lambda}_i^2 \right)^{\frac{1}{2}} &= \sqrt{s_0} \left(\sum_{j=1}^s \lambda_j^2 \right)^{\frac{1}{2}} = \sqrt{s_0} \sigma \left(\sum_{j=1}^s \log \frac{2ed}{s_0} + \frac{2}{s_0} \log \frac{4em}{j} \right)^{\frac{1}{2}} \\ &\leq C \frac{\gamma \sqrt{ns s_0}}{4 + \sqrt{2}} \lambda^\#, \end{aligned} \quad (42)$$

where constant $C \in (1, 2)$. Since the constant term can be ignored, we set C to 1 for convenience. On the other hand, $\tilde{\lambda}_{ss_0+1} = \lambda_{s+1} \leq \frac{\gamma \sqrt{n}}{4 + \sqrt{2}} \lambda^\#$. Combining this inequality, (41) and (42), we have

$$\begin{aligned} (4 + \sqrt{2})N(u) &\leq \gamma \lambda^\# \left\{ 2\sqrt{s s_0} \|u\|_2 + \left(\sum_{i=ss_0+1}^p |u_{(i)}| + \sum_{j=s+1}^m \sqrt{s_0} \|U_{(j)}\|_2 \right) \right\} \\ &\triangleq F(u). \end{aligned}$$

By the definition of $\varrho(\lambda)$ and $\lambda^\#$, we have

$$G(u) = (4 + \sqrt{2}) \|Xu\|_n \sigma \sqrt{\frac{\log \frac{1}{\Delta_0}}{n}} = \lambda^\# \gamma \sqrt{s s_0} \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#)) s s_0} \right)^{\frac{1}{2}} \|Xu\|_n.$$

From Theorem 2.2, with probability at least $1 - \frac{\Delta_0}{2}$, we have

$$\begin{aligned} B_1 &\leq (4 + \sqrt{2}) \max \left(N(u), \|Xu\|_n \sigma \sqrt{\frac{\log \frac{1}{\Delta_0}}{n}} \right) \\ &\leq \max(F(u), G(u)). \end{aligned} \quad (43)$$

Next, for part B_2 , it is worth noting that sequences with all entries equal to $\lambda^\#$ constitute a special case of non-increasing sequences λ and $\tilde{\lambda}$. By Lemma C.1, we have

$$\begin{aligned} B_2 &\leq \lambda^\# \{ (\|\beta^\star\|_1 - \|\hat{\beta}\|_1) + \sqrt{s_0} (\|\beta^\star\|_{1,2} - \|\hat{\beta}\|_{1,2}) \} \\ &\leq \lambda^\# \left\{ \left(\sqrt{s s_0} \|u\|_2 - \sum_{i=ss_0+1}^p |u_{(i)}| \right) + \left(\sqrt{s s_0} \|u\|_2 - \sum_{j=s+1}^m \sqrt{s_0} \|U_{(j)}\|_2 \right) \right\} \\ &\leq \lambda^\# \left\{ 2\sqrt{s s_0} \|u\|_2 - \left(\sum_{i=ss_0+1}^p |u_{(i)}| + \sum_{j=s+1}^m \sqrt{s_0} \|U_{(j)}\|_2 \right) \right\}. \end{aligned} \quad (44)$$

Then, combining parts (43) and (44), we have

$$\begin{aligned} \|Xu\|_n^2 &\leq \max(F(u), G(u)) + \\ &\quad \lambda^\# \left[2\sqrt{s s_0} \|u\|_2 - \left(\sum_{i=ss_0+1}^p |u_{(i)}| + \sum_{j=s+1}^m \sqrt{s_0} \|U_{(j)}\|_2 \right) \right]. \end{aligned} \quad (45)$$

The following proof is discussed in two cases:

1. $G(u) \geq F(u)$. In this case, we have

$$\|u\|_2 \leq \frac{1}{2} \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#))s s_0} \right)^{\frac{1}{2}} \|Xu\|_n. \quad (46)$$

Substituting (46) into (45), we have

$$\|Xu\|_n^2 \leq (1 + \gamma)\lambda^\# \sqrt{s s_0} \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#))s s_0} \right)^{\frac{1}{2}} \|Xu\|_n. \quad (47)$$

Combining (46) and (47), we have

$$\|u\|_2 \leq \frac{1}{2} (1 + \gamma) \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#))s s_0} \right) \sqrt{s s_0} \lambda^\#.$$

2. $F(u) > G(u)$. Under this assumption, (45) shows that

$$\|Xu\|_n^2 \leq \lambda^\# \left[2(1 + \gamma)\sqrt{s s_0} \|u\|_2 - (1 - \gamma) \left(\sum_{i=s s_0+1}^p |u_{(i)}| + \sum_{j=s+1}^m \sqrt{s_0} \|U_{(j)}\|_2 \right) \right]. \quad (48)$$

Therefore, we have

$$\frac{2(1 + \gamma)}{1 - \gamma} \sqrt{s s_0} \|u\|_2 > \sum_{i=s s_0+1}^p |u_{(i)}| + \sum_{j=s+1}^m \sqrt{s_0} \|U_{(j)}\|_2,$$

which implies that u belongs to the cone $C_{SSGRE}(s, s_0, \frac{4\gamma}{1-\gamma})$. Then, according to Assumption 2 with parameter $\theta(s, s_0, \frac{4\gamma}{1-\gamma})$, we have

$$\|u\|_2^2 \leq \frac{\|Xu\|_n^2}{\theta(s, s_0, \frac{4\gamma}{1-\gamma})^2}.$$

Substituting (48) into the above inequality, we have

$$\|u\|_2 \leq \frac{2(1 + \gamma)\sqrt{s s_0} \lambda^\#}{\theta(s, s_0, \frac{4\gamma}{1-\gamma})^2}.$$

Thus, combining these two cases, we conclude that

$$\|\hat{\beta} - \beta^\star\|_2 \leq (1 + \gamma)\sqrt{s s_0} \lambda^\# \max \left\{ \frac{1}{2} \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#))s s_0} \right), \frac{2}{\theta(s, s_0, \frac{4\gamma}{1-\gamma})^2} \right\}. \quad \square$$

C.2. Proof of Theorem 4.1

Proof. Define $\Lambda(s)$ as

$$\Lambda(s) = \frac{(4 + \sqrt{2})}{\sqrt{n}\gamma} \left(\sum_{i=1}^{s s_0} \tilde{\lambda}_i^2 \right)^{\frac{1}{2}} = \frac{(4 + \sqrt{2})}{\sqrt{n}\gamma} \sqrt{s_0} \left(\sum_{j=1}^s \lambda_j^2 \right)^{\frac{1}{2}}.$$

By the definition of $h(\beta)$ and Lemma C.1, we have

$$\begin{aligned} \frac{1}{2} (h(\beta^\star) - h(\hat{\beta})) &= \frac{4 + \sqrt{2}}{\sqrt{n}\gamma} \{ (\|\beta^\star\|_{\tilde{\lambda}^\star} - \|\hat{\beta}\|_{\tilde{\lambda}^\star}) + (\|\beta^\star\|_{G, \lambda^\star} - \|\hat{\beta}\|_{G, \lambda^\star}) \} \\ &\leq 2\Lambda(s)\|u\|_2 - \frac{4 + \sqrt{2}}{\sqrt{n}\gamma} \left(\sum_{i=ss_0+1}^P \tilde{\lambda}_i |u_{(i)}| + \sqrt{s_0} \sum_{j=s+1}^m \lambda_j \|U_{(j)}\|_2 \right). \end{aligned} \quad (49)$$

Define $K(u) = \frac{4 + \sqrt{2}}{\sqrt{n}\gamma} \left(\sum_{i=ss_0+1}^P \tilde{\lambda}_i |u_{(i)}| + \sqrt{s_0} \sum_{j=s+1}^m \lambda_j \|U_{(j)}\|_2 \right)$, then we have

$$\frac{1}{2} (h(\beta^\star) - h(\hat{\beta})) \leq 2\Lambda(s)\|u\|_2 - K(u).$$

According to (10), we have

$$(4 + \sqrt{2})N(u) \leq \gamma \{2\Lambda(s)\|u\|_2 + K(u)\}.$$

Define $H(u) \triangleq \gamma \{2\Lambda(s)\|u\|_2 + K(u)\}$. By Lemma C.2, similar to (45), we get

$$\|Xu\|_n^2 \leq \max\{H(u), G(u)\} + 2\Lambda(s)\|u\|_2 - K(u), \quad (50)$$

where $G(u)$ is defined in the proof of Theorem 3.1. Similarly, we discuss the same in two cases:

1. $H(u) \leq G(u)$. This case implies that

$$2\Lambda(s)\|u\|_2 \leq \lambda^\# \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#))} \right)^{\frac{1}{2}} \|Xu\|_n.$$

By the definition of $\lambda^\#$, we have

$$\|u\|_2 \leq \frac{1}{2} \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#))_{ss_0}} \right)^{\frac{1}{2}} \|Xu\|_n. \quad (51)$$

Substituting the above inequality into (50), we have

$$\|Xu\|_n^2 \leq (1 + \gamma)\lambda^\# \sqrt{ss_0} \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#))_{ss_0}} \right)^{\frac{1}{2}} \|Xu\|_n.$$

Combining these two inequalities, we have

$$\|u\|_2 \leq \frac{1 + \gamma}{2} \left(\frac{\log(1/\Delta_0)}{\log(1/\varrho(\lambda^\#))_{ss_0}} \right) \sqrt{ss_0} \lambda^\#.$$

2. $H(u) > G(u)$. In this case, (50) shows that

$$\|Xu\|_n^2 \leq 2(1 + \gamma)\Lambda(s)\|u\|_2 - (1 - \gamma)K(u), \quad (52)$$

which implies that

$$K(u) < \frac{2(1 + \gamma)}{1 - \gamma} \Lambda(s)\|u\|_2.$$

Consequently,

$$\begin{aligned}\|u\|_{\tilde{\lambda}^*} + \sqrt{s_0}\|u\|_{G, \lambda^*} &= \sum_{j=1}^s \|U_{(j)}\|_2 \sqrt{s_0} \lambda_j + \sum_{i=1}^{ss_0} |u_{(i)}| \tilde{\lambda}_i + K(u) \\ &\leq 2\Lambda(s)\|u\|_2 + K(u) \\ &\leq \left(2 + \frac{2(1+\gamma)}{1-\gamma}\right) \Lambda(s)\|u\|_2.\end{aligned}$$

It shows that u belongs to the cone $C_{WSGRE}(s, s_0, \frac{2(1+\gamma)}{1-\gamma})$. Then, according to Assumption 3, we have

$$\|u\|_2^2 \leq \frac{\|Xu\|_n^2}{\theta(s, s_0, \frac{2(1+\gamma)}{1-\gamma})^2}.$$

Substituting (52) into above inequality and combining (42), we have

$$\|u\|_2 \leq \frac{2(1+\gamma)\Lambda(s)}{\theta(s, s_0, \frac{2(1+\gamma)}{1-\gamma})^2} \leq \frac{2(1+\gamma)\sqrt{ss_0}\lambda^\#}{\theta(s, s_0, \frac{2(1+\gamma)}{1-\gamma})^2}.$$

Combining these two cases, we complete the proof of Theorem 4.1. $\square$

D. Proof of minimax lower bound

D.1. Proof of Lemma 5.2

Proof. With a little abuse of notations, we denote $\theta_{g_j} \in \mathbb{R}^p$ by only preserving the entries of group j , and setting others to 0. In other words, $\theta = \sum_{j=1}^m \theta_{g_j}$.

Rethinking Lemma 5.1, let $\mathcal{M}(s, s_0)$ represent the (s, s_0) -sparse packing set by Lemma 5.1, and $\mathcal{G}(s, s_0)$ the group index set of $\mathcal{M}(s, s_0)$. Therefore $\forall \theta \in \mathcal{M}(s, s_0)$, $|\{j : \theta_{g_j} \neq 0\}| = s$; for $\forall j \in [m]$, we have $\|\theta_{g_j}\|_0 = s_0$. Moreover, the Hamming distance $\rho_H(\theta_i, \theta_k) \geq \frac{ss_0}{4}, \forall i \neq k, \theta_i, \theta_k \in \mathcal{M}(s, s_0)$. Reviewing the construction of the encoding procedure, all entries of θ are 1 or 0.

Denote $G(\theta) = \{j : \theta_{g_j} \neq 0\} \subseteq [m]$. We prove the following result: given any $s \in [\frac{m}{2}]$, and for $\forall G \in \mathcal{G}(s, s_0)$, there exists a vector θ with the following properties:

- $G(\theta) = G$, and $\|\theta_{g_j}\|_0 = s_0, \forall j \in G(\theta)$;
- all the absolute of non-zero entries is $\frac{1}{\sqrt{ss_0}}$;
- $\|X\theta\|_n \leq \vartheta_{\max}$.

The first property holds for Lemma 5.1 and the second is self-evident. We prove the third property by mathematical induction. For $s = 1$, the third property is trivial since Assumption 1 holds. Next, we assume it is right for $s - 1$, and prove that it holds for s . We give a $G \in \mathcal{G}(s, s_0)$, and $G' \subseteq G, |G'| = s - 1$. So that there exists a vector θ with non-zero entries $\frac{1}{\sqrt{ss_0}}$ whose group index $G(\theta) = G'$. We have

$$\|X\theta\|_n^2 \leq \vartheta_{\max}^2 \frac{s-1}{s}.$$

Now consider two vectors θ_1, θ_2 that all non-zero entries are $\frac{1}{\sqrt{ss_0}}$, subject to

$$\begin{cases} (\theta_1)_{g_j} = (\theta_2)_{g_j} = \theta_{g_j} & j \in G' \\ (\theta_1)_{g_j} = -(\theta_2)_{g_j}, & j \in G \setminus G' \\ (\theta_1)_{g_j} = (\theta_2)_{g_j} = 0, & \text{otherwise} \end{cases}$$

Therefore, we have

$$\|X\theta_1\|_n^2 + \|X\theta_2\|_n^2 \leq 2\vartheta_{\max}^2.$$

Consequently, either θ_1 or θ_2 preserves $\|X\theta\|_n \leq \vartheta_{\max}$, which means that these three properties hold for s . This procedure indicates that for any group index vector $G \in \mathcal{G}(s, s_0)$, there exists a vector such that the above properties hold. Therefore, we complete the proof of Lemma 5.2. $\square$

D.2. Proof of Theorem 5.3

Proof. Consider the $\frac{ss_0}{4}$ -packing set $\{\theta^1, \dots, \theta^M\}$ we derive in Lemma 5.2. Let $\beta^{(i)} = \delta\theta^i$, where δ is a parameter that needs to be determined below. For any $\beta^{(i)} \neq \beta^{(j)}$, since $\{\theta^1, \dots, \theta^M\}$ is a $\frac{ss_0}{4}$ -packing set of $\Theta^{m,d}(s, s_0)$, we have

$$\|\beta^{(i)} - \beta^{(j)}\|_2^2 \geq \frac{1}{4}ss_0\delta^2, \quad \forall i, j \in [M]. \quad (53)$$

On the other hand, given design matrix X , we have:

$$\|X\beta^{(i)}\|_n^2 \leq \vartheta_{\max}^2 ss_0\delta^2, \quad \forall i \in [M]. \quad (54)$$

Denote $y^{(i)} = X\beta^{(i)} + \xi_i, \forall i \in [M]$. We consider the Kullback-Leibler divergence between different distribution pairs as

$$\begin{aligned} KL(y^{(i)} \| y^{(j)}) &= \frac{1}{2\sigma^2} \|X(\beta^{(i)} - \beta^{(j)})\|_2^2 \\ &\leq \frac{n}{2\sigma^2} (\|X\beta^{(i)}\|_n^2 + \|X\beta^{(j)}\|_n^2) \\ &\leq \frac{nss_0\vartheta_{\max}^2\delta^2}{\sigma^2}, \end{aligned}$$

where the last inequality uses (54). Denote B as the random vector uniformly distributed over the packing set. Observe that

$$\begin{aligned} I(y; B) &\leq \frac{1}{\binom{M}{2}} \sum_{i < j} KL(y^{(i)} \| y^{(j)}) \\ &\leq \frac{1}{\binom{M}{2}} \sum_{i < j} \frac{1}{2\sigma^2} \|X(\beta^{(i)} - \beta^{(j)})\|_2^2 \\ &\leq \frac{nss_0\vartheta_{\max}^2\delta^2}{\sigma^2}, \end{aligned} \quad (55)$$

Combining the generalized Fano's Lemma and (55), we have

$$\mathbf{P}(B \neq \tilde{\beta}) \geq 1 - \frac{I(y; B) + \log 2}{\log M}$$

$$\geq 1 - \frac{\frac{n\vartheta_{\max}^2}{\sigma^2} ss_0 \delta^2 + \log 2}{\log M},$$

where $\tilde{\beta}$ takes value in the packing set. To guarantee $P(B \neq \tilde{\beta}) \geq \frac{1}{2}$, it suffices to choose $\delta = \frac{1}{2} \sqrt{\frac{\sigma^2}{n\vartheta_{\max}^2 ss_0} \log M}$. Substituting it into equation (53) and from Lemma 5.1, we have

$$\begin{aligned} & \inf_{\tilde{\beta}} \sup_{\beta^* \in \mathcal{DS}(s, s_0)} \mathbf{P} \left(\|\hat{\beta} - \beta^*\|_2^2 \geq \frac{\sigma^2}{64n\vartheta_{\max}^2} (s \log \frac{em}{s} + ss_0 \log \frac{ed}{s_0}) \right) \\ & \geq \inf_{\tilde{\beta}} \sup_{\beta^* \in \Theta^{m,d}(s, s_0)} \mathbf{P} \left(\|\hat{\beta} - \beta^*\|_2^2 \geq \frac{\sigma^2}{64n\vartheta_{\max}^2} (s \log \frac{em}{s} + ss_0 \log \frac{ed}{s_0}) \right) \\ & \geq \mathbf{P}(B \neq \tilde{\beta}) \\ & \geq \frac{1}{2}, \end{aligned}$$

which completes the proof of Theorem 5.3. $\square$

E. Proof of weak random design

E.1. Proof of Theorem 6.1

Proof. Given a vector $\beta \in \mathbb{R}^{d \times m}$, it can be arranged into a group matrix of $d \times m$. where the j -th group is denoted as $\beta_{g_j} \in \mathbb{R}^d$. This proof uses β_{g_j} to denote the parameters of the j -th group, and G denotes the random variables to represent the group index. We divide the random variable construction into four steps as follows:

step 1

Construct the group indicator variable G :

Define

$$w_j \triangleq \sqrt{s_0} \|\beta_{g_j}\|_2 + \|\beta_{g_j}\|_1, \quad \forall j = 1, 2, \dots, m.$$

Then $W = \sum_{j=1}^m w_j$. We define G to obey

$$\mathbf{P}(G = j) = \frac{w_j}{W}, \quad \forall j \in [m].$$

Now let $G_1, G_2, \dots, G_s$ be s random vectors that are identically distributed with G and independent of each other.

step 2

Now given $G_k = j$. Define the conditional random variable Y_G :

$$\mathbf{P} \left(Y_G = \left(\frac{W}{w_j} \right) \cdot \|\beta_{g_j}\|_1 e_{ij} \middle| G = j \right) = \frac{|\beta_{ij}|}{\|\beta_{g_j}\|_1}, \quad \forall i \in [d],$$

where the indicator variable $e_{ij} \in \{0, 1\}^{d \times m}$ means that 1 is taken only at the i -th variable of the j -th group and 0 at other positions.

Similarly assume that $(Y_G)_1, (Y_G)_2, \dots, (Y_G)_{s_0}$ are s_0 random vectors that are identically distributed with Y_G and independent of each other. Then let $H = \frac{1}{s_0} \sum_{l=1}^{s_0} (Y_G)_l$, it is clear that H has elements only on the j -th group and is a s_0 -sparse vector.

For convenience given $G = j$, we also define $\tilde{Y}_G = \frac{w_j}{W} Y_G$, i.e.

$$\mathbf{P}\left(\tilde{Y}_G = \|\beta_{g_j}\|_1 e_{ij} \middle| G = j\right) = \frac{|\beta_{ij}|}{\|\beta_{g_j}\|_1} \quad \forall i \in [d].$$

Then we have:

$$\mathbf{E}\left[\tilde{Y}_G \middle| G = j\right] = \beta_{g_j}.$$

and denote $\tilde{H} = \frac{1}{s_0} \sum_{l=1}^{s_0} (\tilde{Y}_G)_l$.

$$\begin{aligned} \mathbf{E}[\|\tilde{H}\|^2 | G = j] &= \frac{1}{s_0^2} \sum_{l \neq l'} \langle \mathbf{E}[(\tilde{Y}_G)_l | G = j], \mathbf{E}[(\tilde{Y}_G)_{l'} | G = j] \rangle + \frac{1}{s_0^2} \sum_{l=1}^{s_0} \mathbf{E}[\|(\tilde{Y}_G)_l\|^2 | G = j] \\ &= (1 - \frac{1}{s_0}) \|\beta_{g_j}\|_2^2 + \frac{1}{s_0} \sum_{i=1}^d \frac{|\beta_{ij}|}{\|\beta_{g_j}\|_1} \|\beta_{g_j}\|_1^2 \\ &= (1 - \frac{1}{s_0}) \|\beta_{g_j}\|_2^2 + \frac{1}{s_0} \|\beta_{g_j}\|_1^2. \end{aligned} \tag{56}$$

Then for each group of schematic variables given by **step 1**, we can generate the corresponding mutually independent conditional distributions $\mathbf{P}(H_k | G_k)$ respectively. In this way, we obtain s mutually independent random variables $\{H_k\}$.

At this point, we define

$$Z \triangleq \frac{1}{s} \sum_{k=1}^s H_k.$$

Then at this point, Z belongs to the set of doubly sparse vectors $\mathcal{DS}(s, s_0)$. and we have H_{k_1} and H_{k_2} independent of each other at $k_1 \neq k_2$, and:

$$\begin{aligned} \mathbf{E}[H] &= \sum_{j=1}^m \frac{w_j}{W} \mathbf{E}[H | G = j] \\ &= \sum_{j=1}^m \frac{w_j}{W} \mathbf{E}[H | G = j] = \sum_{j=1}^m \frac{w_j}{W} \mathbf{E}[Y_G | G = j] \\ &= \sum_{j=1}^m \frac{w_j}{W} \sum_{i=1}^d \left(\frac{W}{w_j} \right) \cdot \|\beta_{g_j}\|_1 e_{ij} \cdot \frac{|\beta_{ij}|}{\|\beta_{g_j}\|_1} = \beta. \end{aligned}$$

step 3

Thus under the assumptions of the DSRE condition, there are:

$$\mathbf{E}\|XZ\|^2 \geq \theta^2 \mathbf{E}\|Z\|^2.$$

For finding $\mathbf{E}\|XZ\|^2$ we have:

$$\begin{aligned} \mathbf{E}\|XZ\|_2^2 &= \frac{1}{s^2} \sum_{k_1 \neq k_2=1}^s \mathbf{E}\langle XH_{k_1}, XH_{k_2} \rangle + \frac{1}{s^2} \sum_{k=1}^s \mathbf{E}\|XH_k\|^2 \\ &= (1 - \frac{1}{s}) \|X\beta\|_2^2 + \frac{1}{s} \mathbf{E}\|XH\|_2^2. \end{aligned}$$

For the second part:

$$\begin{aligned}
\mathbf{E}\|XH\|_2^2 &= \sum_{j=1}^m \left(\frac{W}{w_j}\right)^2 \mathbf{E} [\|X\tilde{H}\|_2^2 | G = j] \cdot \frac{w_j}{W} \\
&\leq \sum_{j=1}^m \left(\frac{W}{w_j}\right) \mathbf{E} [\|\tilde{H}\|_2^2 | G = j] \\
&= \sum_{j=1}^m \left(\frac{W}{w_j}\right) \left[\left(1 - \frac{1}{s_0}\right) \|\beta_{g_j}\|_2^2 + \frac{1}{s_0} \|\beta_{g_j}\|_1^2 \right] \\
&\leq \frac{1}{s_0} \sum_{j=1}^m \left(\frac{W}{w_j}\right) [\sqrt{s_0} \|\beta_{g_j}\|_2 + \|\beta_{g_j}\|_1]^2 \\
&= \frac{1}{s_0} \sum_{j=1}^m \left(\frac{W}{w_j}\right) w_j^2 = \frac{1}{s_0} W^2.
\end{aligned} \tag{57}$$

The first inequality holds because $\tilde{H}$ is a s_0 sparse vector over a single group, so according to the sparse group normalization Assumption 1, we have $\|X\tilde{H}\|_2 \leq \|\tilde{H}\|_2$. So that:

$$\mathbf{E}\|XZ\|_2^2 \leq \left(1 - \frac{1}{s}\right) \|X\beta\|_2^2 + \frac{1}{ss_0} W^2. \tag{58}$$

Similarly, we derive:

$$\mathbf{E}\|Z\|_2^2 = \left(1 - \frac{1}{s}\right) \|\beta\|_2^2 + \frac{1}{ss_0} W \sum_{j=1}^m \frac{1}{w_j} [(s_0 - 1) \|\beta_{g_j}\|_2^2 + \|\beta_{g_j}\|_1^2]. \tag{59}$$

Combining (58), (59) yields:

$$\|X\beta\|_2^2 \geq \theta^2 \|\beta\|_2^2 - \frac{1}{s_0(s-1)} W^2.$$

Thus when β satisfies $W^2 \leq \frac{s_0(s-1)}{2} \theta^2 \|\beta\|_2^2$, we have $\|X\beta\|_2^2 \geq \frac{\theta^2}{2} \|\beta\|_2^2$. Thus when X satisfies the double sparse RE condition $DSRE(s, s_0, \theta)$, we can let $\tilde{\theta} = \frac{\theta}{\sqrt{2}}$ and $\tilde{s} = \frac{(s-1)\theta^2}{2(2+c_0)^2}$, then X satisfies the strong sparse group RE condition $SSGRE(\tilde{s}, s_0, c_0, \tilde{\theta})$. $\square$

E.2. Proof of Theorem 6.2

Here we first derive the VC-dimension of the class $\mathcal{F}(s, s_0) = \{f(x) = x^\top \beta : \beta \in \mathcal{DS}(s, s_0)\}$, then apply the result of corollary 2.5 of [22], which derives Theorem 6.2 directly.

For brevity, V denotes the VC-dimension of $\mathcal{F}(s, s_0)$. By the definition of VC-dimension, For any fixed ss_0 -dimension, the VC-dimension of the corresponding set ss_0 -dimensional linear classifiers is known to be $ss_0 - 1$.

Then, by Sauer's lemma (e.g., [46], Proposition 4.18) the maximal number of different labeling of V points in R^{d_0} that such set of classifiers can produce is upper bounded by $\sum_{k=0}^{ss_0} \binom{V}{k}$. So considering a union of $\binom{m}{s} \binom{sd}{ss_0}$'s ss_0 -dimensional linear classification, we have

$$2^V \leq \binom{m}{s} \binom{sd}{ss_0} \sum_{i=0}^{ss_0} \binom{V}{i}$$

$$\leq \left(\frac{em}{s}\right)^s \left(\frac{ed}{s_0}\right)^{ss_0} \left(\frac{eV}{ss_0}\right)^{ss_0}.$$

Denote $Z = \frac{V}{ss_0}$. By some simple algebra, we have

$$2^Z \leq \left(\frac{em}{s}\right)^s \left(\frac{ed}{s_0}\right)^{ss_0} (eZ)^{ss_0}. \quad (60)$$

We need to find the largest integer that satisfies (60). To guarantee this, it is sufficient to choose $Z \leq 2\left(\frac{1}{s_0} \log \frac{em}{s} + \log \frac{ed}{s_0}\right)$ as an upper bound of Z . Consequently, we derive the upper bound $V \leq 2(s \log \frac{em}{s} + ss_0 \log \frac{ed}{s_0})$.

E.3. Proof of weak moment normalization condition

Lemma E.1 (Lemma 2.8 in [22]). *There exists an absolute constant c_0 for which the following holds. Let Z be a mean-zero random variable, and let $Z_1, Z_2, \dots, Z_n$ be n independent copies of Z . Let $q_0 \geq 2$ and assume that there exist $\kappa_1 > 0$ and $\alpha > \frac{1}{2}$, for which $\|Z\|_{L_q} \leq \kappa_1 q^\alpha$, $\forall 2 \leq q \leq q_0$. If $n \geq q_0^{\max(2\alpha-1, 1)}$, then for every $2 \leq q \leq q_0$, we have*

$$\left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i \right\|_{L_q} \leq c_1(\alpha) \kappa_1 \sqrt{q}, \quad c_1(\alpha) = \exp(2\alpha - 1).$$

proof of Theorem 6.3. Given any $j \in \mathcal{S}$, we define $Z = (X_j^*)^2 - \|X_j^*\|_{L_2}^2$. So that $E[Z] = 0$ and we have $\|Z\|_{L_q}^q \leq \|X_j^*\|_{2q}^{2q} \leq \kappa_1^{2q} (2q)^{2\alpha q}$, $\forall 2 \leq q \leq q_0$, where q_0 will be chosen later. By Lemma E.1, if $n \geq q_0^{\max(4\alpha-1, 1)}$, we have:

$$\left\| \frac{1}{n} \sum_{i=1}^n (X_{ij}^*)^2 \right\|_{L_q} \leq 1 + 4\alpha c_1(2\alpha) \kappa_1^2 \sqrt{\frac{q}{n}}, \quad \forall 2 \leq q \leq q_0. \quad (61)$$

For α, κ_1 are absolute constants, we obtain the second condition $n \geq [4\alpha c_1(2\alpha) \kappa_1^2]^2 q \asymp q$, and for any j , we have $\left\| \frac{1}{n} \sum_{i=1}^n (X_{ij}^*)^2 \right\|_{L_q} \leq 2$. On the other hand, we use Markov's inequality and union bound by the cardinal $|\mathcal{S}| = m \binom{d}{s_0}$:

$$\mathbf{P} \left(\max_{j \in \mathcal{S}} \frac{1}{n} \sum_{i=1}^n (X_{ij}^*)^2 \geq 2e \right) \leq |\mathcal{S}| \left(\frac{\left\| \frac{1}{n} \sum_{i=1}^n (X_{ij}^*)^2 \right\|_{L_q}}{2e} \right)^q \leq \exp(\log |\mathcal{S}| - q).$$

Therefore, we obtain our conclusion by combining the three conditions above. If there exists a constant $C_1 > 1$ that $q_0 \geq q > C_1(\log m + s_0 \log \frac{ed}{s_0})$ holds, and sample size $n \geq q_0^{\max(4\alpha-1, 1)}$, then the sparse group normalization assumption 1 holds with probability larger than $1 - \exp\{-(C_2 - 1)(\log m + s_0 \log \frac{ed}{s_0})\}$. $\square$

F. Proof of sub-Gaussian random design

F.1. Preliminaries

At first, we define ϵ -cover as follows:

Definition F.1 (ϵ -cover). Given a set $\mathcal{T}$ and a semimetric ρ , there exists a set $\{\theta^1, \theta^2, \dots, \theta^N\} \subset \mathcal{T}$. For any $\theta \in \mathcal{T}$, there exists some $i \in [N]$ such that $\rho(\theta, \theta^i) \leq \epsilon$. We call this set the ϵ -cover of $\mathcal{T}$. The ϵ -covering number $N(\epsilon; \mathcal{T}, \rho)$ is the cardinality of the smallest ϵ -cover of $\mathcal{T}$.

Referring to the steps of [52], we need the following lemma from [32], which we give without proof.

Lemma F.2 ([32], Lemma 2.2). Given $m \geq 1$ and $\epsilon > 0$. There exists an ϵ -cover $\Pi \subseteq B_2^m$ of B_2^m with respect to the Euclidean metric such that $B_2^m \subseteq (1 - \epsilon)^{-1} \text{conv}(\Pi)$ and $|\Pi| \leq (1 + 2/\epsilon)^m$. Similarly, there exists an ϵ -cover $\Pi' \subseteq S^{m-1}$ of the sphere S^{m-1} such that $|\Pi'| \leq (1 + 2/\epsilon)^m$.

Note that the cardinality of $\mathcal{S}$ is $|\mathcal{S}| = m \cdot \binom{d}{s_0}$. We provide the lemma corresponding to the ϵ -cover of V_{s_0} and $\tilde{V}_{s_0}$ as follows

Lemma F.3. For any given $0 < \epsilon < \frac{1}{2}$ and $0 < s_0 < d$, there exists an ϵ -cover set $\Pi \subseteq B_2^p$ of $\tilde{V}_{s_0}$ such that $\tilde{V}_{s_0} \subseteq 2\text{conv}(\Pi)$ and $|\Pi|$ is at most

$$|\Pi| \leq m \cdot \binom{d}{s_0} \left(\frac{5}{2\epsilon} \right)^{s_0}. \quad (62)$$

Moreover, since $V_{s_0} \subseteq \tilde{V}_{s_0}$, Π is also an ϵ -cover set of V_{s_0} .

Proof. The proof is the same as Lemma 2.3 of [32] and Lemma B.4 of [52]. Just need to replace the number of unit balls according to $|\mathcal{S}|$. $\square$

Finally, we need to use the following lemma:

Lemma F.4 ([23]). Let $X_1, X_2, \dots, X_N$ be p -dimensional Gaussian random vectors. Then,

$$\mathbf{E} \max_{i=1,2,\dots,N} |X_i| \leq 3\sqrt{\log N} \max_{i=1,2,\dots,N} \sqrt{\mathbf{E}\|X_i\|_2^2}.$$

F.2. Proof of Theorem 6.7

Proof. Let g be a p -dimensional random vector with each entry drawn from $\mathcal{N}(0, 1)$ independently. Since $V_{s_0} \subseteq 2\text{conv}(\Pi)$ and according to the convexity, we have

$$\sup_{v \in \text{conv}(\Pi)} \langle v, \Sigma^{\frac{1}{2}} g \rangle = \sup_{v \in \Pi} \langle v, \Sigma^{\frac{1}{2}} g \rangle.$$

Therefore,

$$\begin{aligned} \tilde{\ell}^*(V_{s_0}) &\leq \mathbf{E} \sup_{v \in 2\text{conv}(\Pi)} \langle v, \Sigma^{\frac{1}{2}} g \rangle \\ &= 2\mathbf{E} \sup_{v \in \Pi} \langle v, \Sigma^{\frac{1}{2}} g \rangle \\ &\leq 6\sqrt{\log |\Pi|} \max_{S \in \mathcal{S}} \sqrt{\mathbf{E}|\langle v_S, \Sigma^{\frac{1}{2}} g \rangle|^2} \\ &\leq 6 \left(\log m + s_0 \log \frac{5ed}{s_0} \right)^{\frac{1}{2}}. \end{aligned} \quad (63)$$

We plug (63) into the Lemma 6.6, when satisfying the formula (24), for $\forall v \in V_{s_0}$, S is recorded as the support set of v , there are:

$$\frac{\|Xv\|_2}{\sqrt{n}\|v\|_2} = \frac{\|Xv\|_2}{\sqrt{n}\|\Sigma_S^{\frac{1}{2}}v\|_2} \cdot \frac{\|\Sigma_S^{\frac{1}{2}}v\|_2}{\|v\|_2} \leq (1+\theta) \cdot \frac{1}{(1+\theta)} = 1. \quad \square$$

F.3. Proof of Theorem 6.8

Proof. According to Lemma 6.6, we need to find a certain parameter space V , and derive the Gaussian complexity $\tilde{\ell}^*(V)$. We consider the parameter space as follows:

$$V_2 = C_{WSGRE}(s, s_0, c_0) \cap \{v, \|\Sigma^{\frac{1}{2}}v\|_2 = 1\}.$$

So to define $\tilde{V}_2$, let's assume $\sigma = 1$:

$$\tilde{V}_2 \triangleq \left\{ v : v \in \mathbb{R}^p : \|v\|_* \leq \frac{(2+c_0)}{\theta} \sqrt{\sum_{i=1}^{ss_0} \tilde{\lambda}_i^2}, c_0 > 0, \|\Sigma^{\frac{1}{2}}u\|_2 \leq 1 \right\}, \quad (64)$$

where we can derive

$$\left(\sum_{i=1}^{ss_0} \tilde{\lambda}_i^2 \right)^{\frac{1}{2}} = \sqrt{ss_0 \log \frac{2ed}{s_0} + 2s \log \frac{4em}{s}}. \quad (65)$$

Then $V_2 \subseteq \tilde{V}_2$. Define $r \triangleq \frac{(2+c_0)}{\kappa} \sqrt{ss_0 \log \frac{2ed}{s_0} + 2s \log \frac{4em}{s}}$. Because Σ satisfies the sparse group normalization condition, that is, for $\forall S \in \mathcal{S}$ there is $\|\Sigma_S\| \leq 1$, so according to Lemma B.3 and Proposition B.4, since $\|v\|_* \equiv \sqrt{n}N(v)$, we have:

$$\text{Med} \left(\sup_{v: \|v\|_* \leq 1} \left| \langle \xi, \Sigma^{\frac{1}{2}}v \rangle \right| \right) \leq 4. \quad (66)$$

Remark 8. Combining the conclusions in the Lemma B.3 and Proposition B.4 and the positive homogeneity of $N(u)$, we can obtain:

$$\text{Med} \left(\sup_{v: N(v) \leq \frac{1}{\sqrt{n}}} \left| \langle \xi, \frac{1}{\sqrt{n}}Xv \rangle \right| \right) \leq 4.$$

So the proof of (66) is similar, only need to replace $\frac{1}{\sqrt{n}}X$ with $\Sigma^{\frac{1}{2}}$ in the corresponding conditions and conclusions.

Therefore for the parameter space $\tilde{V}_2$, we have:

$$\text{Med} \left(\sup_{v \in \tilde{V}_2} \left| \langle \xi, \Sigma^{\frac{1}{2}}v \rangle \right| \right) \leq 4r,$$

and because there is $\forall v \in \tilde{V}_2, \|\Sigma^{\frac{1}{2}}v\| \leq 1$ in $\tilde{V}_2$, so $f(\xi) = \sup_{v: \|v\|_* \in \tilde{V}_2} \left| \langle \xi, \Sigma^{\frac{1}{2}}v \rangle \right|$ is a 1-Lipschitz function. According to Lemma A.3 of [4]:

$$\mathbf{E} \left(\sup_{v \in \tilde{V}_2} \left| \langle \xi, \Sigma^{\frac{1}{2}}v \rangle \right| \right) \leq 4r + \sqrt{\frac{2}{\pi}}. \quad (67)$$

Plugging (67) into the Lemma 6.6, we make the conclusion as follows: if there is a constant $c', \theta > 0$ such that the sample size n satisfies:

$$n > c' \alpha^4 \theta^2 \left(s s_0 \log \frac{2ed}{s_0} + s \log \frac{4em}{s} \right),$$

then there exists $\bar{c}$, with probability at least $1 - \exp(-\bar{c}\theta^2 n / \alpha^4)$, for all $v \in C_{WSGRE}(s, s_0, c_0) \setminus \{0\}$, according to homogeneity we have

$$\frac{\|Xv\|_n}{\sqrt{n} \|\Sigma^{\frac{1}{2}} v\|_2} \geq (1 - \theta),$$

so that

$$\frac{\|Xv\|_n}{\sqrt{n} \|v\|_2} \geq (1 - \theta) \kappa. \quad \square$$

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